

Routing the Silent Half: Per-Unit Choice of Estimator and Evolution Target in Self-Evolving LLM Agents

Abstract

Group-relative policy optimisation assigns zero advantage to any group whose samples score alike, so a unanimous group contributes exactly nothing to the update. We measure this directly: 57.4% of groups in a live run are RL-silent. We show the silent half splits into two regimes that require opposite responses, and that the split is not symmetric — on the solved half a supervised target is *free*, because the group already contains a correct sample; on the unsolved half no target exists unless one is supplied, which makes harness evolution the only consumer that can use those units at all. We therefore make the choice of estimator and the choice of evolution target a per-unit decision rather than a per-run schedule, with the two as orthogonal axes so a single trajectory can feed both. The closest prior work, DAPO’s dynamic sampling, acts on exactly this set and responds in the opposite way — it discards the silent groups and oversamples to refill — which on our own measurements costs $1.5\text{--}2.4\times$ the rollouts, a multiplier that *grows* with training because the silent fraction itself grows from 0.33 to 0.58 on an arm with no routing at all. We report the negative result that located this leverage: a token-level version of the same idea, reaching 1.7% of tokens, moved held-out accuracy by less than the noise floor. Every capability is flag-gated and reduces to unmodified GRPO when disabled.

1 Method

1.1 Where the gradient is zero

Let a *unit* be a prompt with G sampled answers and binary rewards $r_1, \dots, r_G \in \{0, 1\}$. GRPO centres rewards within the group,

$$A_i = r_i - \bar{r}, \quad \bar{r} = \frac{1}{G} \sum_{j=1}^G r_j, \quad (1)$$

so if every sample in the group scores alike then $A_i = 0$ for all i and the unit contributes exactly nothing to the update — all G sequences, every token. Writing p for the per-sample success probability, the probability that a unit carries any signal at all is

$$I_{\text{RL}}(p, G) = 1 - p^G - (1 - p)^G, \quad (2)$$

which is zero exactly at $p = 0$ and $p = 1$ and is maximised at $p = \frac{1}{2}$. Equation (2) is not an approximation: it is the probability that the group is not unanimous, and unanimity is precisely the condition under which (1) vanishes.

Two consequences drive the method.

The silent fraction is large and measurable. Instrumenting a live run (Sec. 3) puts $\hat{P}(\text{silent})$ between 0.29 and 0.57 depending on the training step, at 64 groups per batch: a third to a half of every batch is RL-dead. Even the low end is an order of magnitude above the reach of the token-level shared-prefix channel we measured first (1.7%), and that gap explains, without appealing to noise, why an intervention confined to that channel moved nothing.

The two silent regimes need opposite responses, and they are not equally common. A unit silent because $\hat{p} = 1$ and a unit silent because $\hat{p} = 0$ are both invisible to (1), but they are not the same object, and routing them identically wastes one of them. Measured on the control arm, the silent channel is 87.5% solved (Table 4), which decides which of the two branches carries the method.

The composition is a property of the task *and* the model, and it moves in both directions. The 87.5% is GSM8K with Qwen2.5-1.5B-Instruct; on MATH, holding the model fixed, it inverts to 39.1% solved and 60.9% unsolved (Table 6). Re-measured on the arm this paper trains — decontaminated DeepMath with Qwen2.5-32B-Instruct, over 245 logged steps of the baseline — the channel is 0.505 of all groups and splits 0.338 solved, 0.167 unsolved and 0.000 unclassified, i.e. 66.9% solved and 33.1% unsolved. The unsolved share is therefore roughly *half* what the MATH column reports, and we state the confound plainly: that pair varies the model as well as the task, so it cannot separate a task effect from a scale effect, and both readings shrink the branch. Two consequences are load bearing. First, any statement sized on the 60.9% figure must not be carried onto this arm: the unsolved branch is about a sixth of every batch here, not a third. Second, the 0.000 unclassified mass means no group on this arm is silent for a reason the correctness split cannot express, so unlike on GSM8K the composition ratio has a clean denominator and unsolved/silent and unsolved/(solved + unsolved) coincide.

1.2 A free target exists on exactly the solved half

The asymmetry is sharper than "different regimes". For a unit with $\hat{p} > 0$, at least one sampled answer was correct, and that answer *is* a supervised target drawn from the model's own rollout. No external teacher is required; this is rejection-sampling fine-tuning. Define

$$\text{selfTarget}(u) = \mathbb{I}[\hat{p}_u > 0], \quad \text{target}(u) = \text{selfTarget}(u) \vee \text{teacher}(u). \quad (3)$$

The units on which $I_{\text{RL}} = 0$ because $\hat{p} = 1$ are therefore exactly the units on which a target is *guaranteed* to exist at zero marginal cost. The units on which $I_{\text{RL}} = 0$ because $\hat{p} = 0$ are exactly the units on which no target exists unless one is supplied. The gradient-free half of the batch splits cleanly into a half that is free to reuse and a half that is not reusable at all.

1.3 Supplying a target for the half that has none

"Not reusable at all" is a statement about (1), not about the batch. An advantage is a coefficient on tokens the model *emitted*, so on a unit with $\hat{p} = 0$ every value that could be written into the advantage tensor multiplies a wrong derivation, and a positive constant there is a step toward the error — which is why $c_- \leq 0$ is required and why no router in our registry reaches this branch. The only altitude at which a correct row can receive gradient is *batch construction*: put the correct response in as a row, with its own prompt and its own loss mask, and the estimator that already runs on every other row runs on it too. This is the construction of LUFFY and of DyME, and it is what makes the unsolved branch addressable at all.

The supplier is an axis, not a constant. Once the row enters by construction, where it came from is a separate question from where it goes: the splice, the masks, the counters and the off-policy treatment are properties of inserting a correct row and do not depend on its origin. We therefore define a supplier interface with four implementations — the dataset’s gold solution; a *self-generated* correct rollout of the same prompt from an earlier step or a higher-temperature resample; a row from an offline *corpus* of solved examples; and a verified completion from a stronger *teacher* — and route a unit to one of them. Prompt identity is derived from the prompt tokens rather than from a batch-local index, because a self-generated row is by definition one the same prompt received at a different step.

A supplier that has nothing must refuse, and the refusal is the measurement. Each supplier either returns token ids or raises a typed refusal, and every reason has its own counter that is reported whether or not it is zero: a missing dataset solution, a prompt never solved before, a derivation too long for the row, and a teacher completion that fails verification have four different remedies, and a single “skipped” total sends the reader to the wrong one. No supplier substitutes a row belonging to a different prompt, and no supplier truncates: a cut-off derivation is a wrong target that still looks like a target. The teacher requires an explicit verifier for the same reason — a stronger model is not a correct one, and this is precisely the branch on which nothing else would catch the error.

The source decision is fixed, and its control is not optional. The choice of supplier is expressible per unit, but we do not learn it here. We use an explicit ordered policy, cheapest-and-most-trusted first, and compare it against a control that replays the policy’s own realised source assignment permuted across units, so the mixture of sources is matched exactly by construction and only the *targeting* differs. This is the same falsification the routing control performs one axis over, and we run it for the same reason: in this project a learned mechanism has repeatedly matched a feature-blind control at matched proportions once the control was finally run, and a claim about choosing sources that has not been separated from a claim about mixing them is not a claim.

1.4 Routing: the harness is a destination, not a schedule

Prior work treats the choice of what to evolve as a property of the *run*: a schedule or a global flag. We make it a property of the *unit*, and we make it two-dimensional. A routing decision is a pair

$$\pi(u) = (m(u), h(u)), \quad m \in \{\text{RL}, \text{SFT}, \text{SKIP}\}, \quad h \in \{\text{NONE}, \text{PROPOSE}, \text{VALIDATE}\}, \quad (4)$$

where m names the estimator applied to the unit’s tokens and h names what the unit contributes to harness evolution. The two axes are orthogonal by construction, so a single trajectory can feed the gradient and the harness at once. This is deliberate: the two consumers read different things from a trajectory — the estimator reads the tokens, the harness evolver reads the failure mode — and a partition discards one of the readings. Co-Harness’s success→model / failure→harness rule is the special case in which π is constrained to be a partition, and we retain it as an ablation (**partition=True**) rather than a rewrite.

The rule implied by (2) and (3) is then forced rather than chosen:

$$\pi(u) = \begin{cases} (\text{SFT}_{\text{self}}, \text{VALIDATE}) & \hat{p}_u = 1 \quad (\text{RL dead; target free}) \\ (\text{SFT}_{\text{teacher}}, \text{PROPOSE}) & \hat{p}_u = 0 \wedge \text{teacher}(u) \\ (\text{SKIP}, \text{PROPOSE}) & \hat{p}_u = 0 \wedge \neg \text{teacher}(u) \quad (\text{harness is the only consumer}) \\ (\text{RL}, \text{NONE}) & 0 < \hat{p}_u < 1 \quad (\text{RL has signal; leave it alone}) \end{cases} \quad (5)$$

Note the third case. When a unit is unsolved and no teacher is available, the model arm has nothing to learn from it, and any method that only evolves weights must discard the unit. Routing it to the harness is what makes it recoverable at all, which is the concrete sense in which the harness axis is not a convenience.

One trajectory, two consumers. (5) as written still couples the two axes: a unit that RL can learn from contributes nothing to the harness. That is a residual partition, and it discards information — a *mixed* unit contains failed samples, and their failure mode is exactly what a harness evolver reads, while the RL update consumes only the reward. We therefore decouple the harness action with its own threshold τ_h :

$$h(u) = \begin{cases} \text{VALIDATE} & \hat{p}_u = 1 \\ \text{PROPOSE} & \hat{p}_u \leq \tau_h \\ \text{NONE} & \text{otherwise,} \end{cases} \quad \tau_h \geq \tau_{\text{fail}}, \quad (6)$$

so at $\tau_h = \tau_{\text{fail}}$ the rule is unchanged, and raising it lets a unit take (RL, PROPOSE) — feeding the gradient and the harness from the same trajectory. Under **partition** this is disabled, so the Co-Harness special case is preserved exactly.

Rollback. Every capability above is gated. With the harness arm disabled, $h(u) \equiv \text{NONE}$; with routing disabled, $\pi(u) \equiv (\text{RL}, \text{NONE})$ for every unit and the update is bit-identical to unmodified GRPO; with the supplier axis of Sec. 1.3 disabled, the batch construction reduces to its single original source and its output is byte-identical, key set included, which we verify against a digest recorded before the generalisation existed rather than by inspection. Each ablation in Sec. 3 is one flag away from the vanilla baseline, so a difference cannot be attributed to an incidental change of configuration.

1.5 Who decides: from a threshold to a controller

Everything above specifies an action space. What chooses within it is a separate question, and it is where the contribution lies — a rule keyed on $\hat{p}$ is a heuristic, and a learned policy that sees only $\hat{p}$ cannot beat the best threshold on $\hat{p}$ by more than noise. We therefore give the controller features, and instantiate it three ways so the comparison is an ablation rather than a demonstration.

Features, at zero marginal cost. Each unit is described by seven statistics computed from tensors the batch *already carries* — rewards, the loss mask, and the sampled log-probabilities. No extra forward pass, no extra generation. The constraint is deliberate: a feature set needing its own inference budget would have to beat spending that budget on more rollouts, which is a far higher bar than beating a threshold. Two of the seven carry information $\hat{p}$ cannot: the *truncated fraction* separates a unit that failed from one that merely ran out of budget, and the *log-probability dispersion* separates a group unanimous in its answer from one also unanimous in its reasoning. A solve-rate threshold collapses both distinctions by construction.

Three controllers, one action space.

- **Rule** — thresholds on $\hat{p}$, as in (5). The baseline the others must beat, not a strawman: it encodes the exact zero-gradient condition.

Algorithm 1 Per-batch signal routing. $\mathcal{B}$ is a batch of prompts, G samples each.

Require: thresholds $\tau_{\text{solve}}=1$, $\tau_{\text{fail}}=0$, $\tau_h \geq \tau_{\text{fail}}$; weights $c_+ \geq 0$, $c_- \leq 0$; flags **routing**, **harness**

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1: roll out  $G$  samples per prompt; score to binary rewards  $r_{u,i}$ 
2:  $\hat{p}_u \leftarrow \frac{1}{G} \sum_i r_{u,i}$  ▷ group solve rate
3:  $A_{u,i} \leftarrow r_{u,i} - \hat{p}_u$  ▷ group-centred advantages, (1)
4:  $\mathcal{S} \leftarrow \{u : \max_i |A_{u,i}| = 0\}$  ▷ RL-silent: exactly the unanimous units
5: if routing then
6:   for all  $u \in \mathcal{S}$  do
7:     if  $\hat{p}_u \geq \tau_{\text{solve}}$  then ▷ solved: a correct sample is its own target
8:        $A_{u,i} \leftarrow A_{u,i} + c_+$  for all response tokens
9:     else if  $\hat{p}_u \leq \tau_{\text{fail}}$  then ▷ failed: no self-target exists
10:       $A_{u,i} \leftarrow A_{u,i} + c_-$  if  $c_- \neq 0$ , else leave at 0
11:    end if
12:  end for
13: end if
14: if harness then
15:    $\mathcal{P} \leftarrow \{u : \hat{p}_u \leq \tau_h\}$ ;  $\mathcal{V} \leftarrow \{u : \hat{p}_u = 1\}$ 
16:   emit (PROPOSE,  $u$ ) for  $u \in \mathcal{P}$  and (VALIDATE,  $u$ ) for  $u \in \mathcal{V}$ 
17: end if
18: PPO update on  $\{A_{u,i}\}$ , clipping unchanged

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- **Learned weights** — LinUCB, one ridge model per mode over the features, updated from observed outcomes. Ridge rather than a network because the outcome channel is confounded and low-signal — one scalar per batch, produced by every decision in it — and capacity would mostly fit that noise.
- **Learned code** — the policy is generated *source*, validated against an allowlist and cost-vetted in a subprocess before adoption. Strictly larger hypothesis class than weights: it can compose features into ratios and nested cases rather than only reweighting them.

The control that decides the claim. A fourth arm shuffles which units receive which mode while holding the *proportion* of units in each mode fixed. Without it, a controller that helps cannot be distinguished from a change in how much SFT the run did — and the latter is not a controller at all. Any improvement reported for the learned arms is the improvement over this control, not over the fixed rule alone.

What we verify about the learned arm, and what we do not. On a synthetic environment whose two regions differ *only* in the truncated fraction, with $\hat{p}$ held identical, the contextual controller reaches the optimal mode in both regions across seeds, while a context-free bandit on the same stream provably cannot exceed 0.5 in the worse region. That establishes the mechanism can use a feature a threshold cannot see. It does *not* establish that it learns under the confounded batch-level channel of a real run, which is a separate claim and is not made here.

1.6 The procedure, end to end

Algorithm 1 is the whole method. Every line is either an existing GRPO step or a decision from (5)–(6); nothing else is added, and with routing disabled lines 6–13 do not execute.

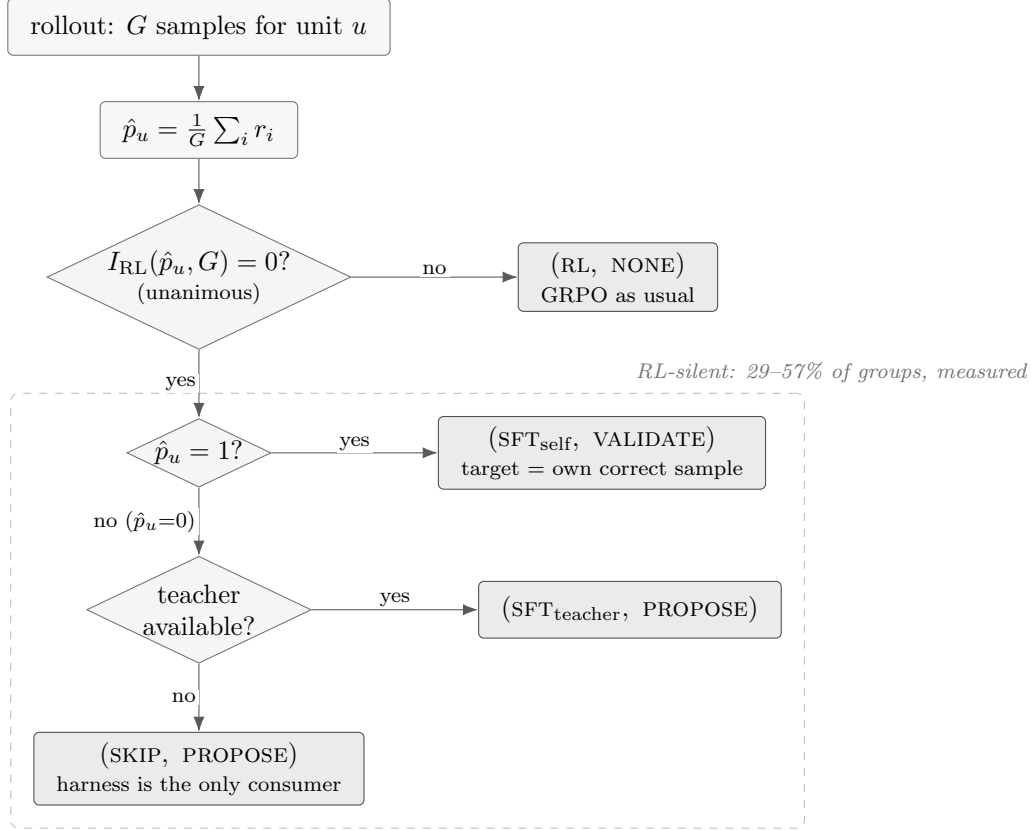


Figure 1: **Routing decides, per unit, both an estimator and a harness contribution.** The branch on I_{RL} is exact, not heuristic: a unanimous group has every advantage identically zero by (1). The dashed region is the part of the batch ordinary GRPO cannot learn from, measured at 29–57% of groups depending on training step. It does not split evenly: 87.5% of it is solved, where a supervised target is free (the unit’s own correct sample), and the remainder has no target unless one is supplied — and on that remainder the harness is the only consumer that can use the unit at all.

Why the update needs no new loss term. Line 8 is the whole of the supervised branch. For a solved unit every $A_{u,i}$ is zero, so adding c_+ makes the objective $c_+ \sum_i \log \pi(y_{u,i})$ over samples already known to be correct — which is supervised fine-tuning on them, with the importance ratio at 1 because the data is on-policy. PPO’s clip still bounds the step, which matters because sharpening an already-correct policy is the mechanism’s predicted failure. Line 10 is its mirror: $c_- \leq 0$ pushes down samples known to be wrong, and $c_- > 0$ is rejected rather than trusted, since it would train the model to reproduce them.

Cost. Lines 5–13 add no generation. The batch is the batch that GRPO already rolled out; the silent units are re-read, not re-sampled. This is the whole of the difference from DAPO, which reaches the same set by discarding it and paying to generate a replacement.

2 Related work

We organise prior work by the claim it already owns, because our contribution is only meaningful as the complement of those claims.

Discarding the silent groups. DAPO’s dynamic sampling is the closest prior work, and it is close in the strongest sense: it acts on exactly the set we act on. Its rule keeps a prompt only when the group’s reward standard deviation is positive, and a zero standard deviation is unanimity, which by (1) is precisely the zero-advantage condition. DAPO therefore *detects* the silent channel and *discards* it, oversampling until the batch refills. We reuse it. Same phenomenon, opposite response — so DAPO is the baseline this claim must be measured against rather than a method to cite in passing, and Sec. 3.6 reports what its response costs.

Choosing what to update. SIA is the nearest work on the *decision* axis, and our earlier reading of it was wrong in a way worth stating rather than silently fixing: it is not task-level algorithm selection. It is a feedback agent that updates *both* the harness and the weights, it is built on a 120B base, and it reports no mathematics or code benchmark at all. Two consequences follow. Its base is outside what our stack can train, and its evaluation domains do not intersect ours, so a head-to-head on its own benchmark would confound the method with the base model and with the domain. The defensible comparison is its rule reimplemented on our base at matched budget, and that is how we report it. The granularity argument itself is unaffected: what a task-level assignment cannot express is that units *within* one task differ in whether a gradient exists at all, since by (2) the silent and informative units of a single task require different estimators.

Deciding what to evolve. Co-Harness co-evolves harness and weights and establishes the success→model / failure→harness assignment. EvoTrainer co-evolves a policy with its training harness. Both fix the assignment as a rule over outcomes. We keep their rule as a special case ((4) constrained to a partition) and relax it, because a trajectory read by two different consumers need not be spent on only one of them.

Evolving the reward itself, and the anchor that is missing. Two recent systems bracket this question and, between them, leave the interesting case open. BigBang-v1 closes a generator–critic loop in which a code agent rewrites the *data-synthesis program* while a critic filters what it produces, and a meta-critic corrects the critic against the training outcome it failed to predict; the loop is explicitly anchored on held-out real-task utility. It has no reward function anywhere, and no reinforcement learning. Ornith-1.5 is its mirror image: the harness it generates per task *is* the reward function, and that harness is itself optimised by policy gradient against a score that includes a hack-resistance term. An evolving reward is therefore already shipped, and we do not claim it. What neither provides is the combination. BigBang has the anchor and nothing that moves; Ornith moves the reward and states no external anchor inside its loop. Its own predecessor held the verifier immutable and outside the model’s reach, and the successor — which now *generates* the object that grades it — does not say what replaced that boundary, while its hack-resistance term is named without being given a form or a scorer. An evolving reward anchored on a fixed external utility is thus unoccupied, and it is the claim we make.

What neither system decides. Neither has a router. BigBang’s critic has a binary codomain — a candidate is admitted, revised, or dropped — and nothing about *how* to learn from an admitted sample varies: one loss and one source throughout. Ornith operates at whole-task granularity, which is the finest thing it does, and what varies per task is artifact content rather than the learning mechanism. It states that per-category strategy emerges rather than being chosen, and it does not test that claim against a frozen or randomised scaffold, which is precisely the absorption control we impose on ourselves and therefore run against our own arms. Neither system conditions on

behavioural clustering features or on trajectory statistics, which is where our decision variable is read.

Deciding *when* to evolve. *Learning, Fast and Slow* separates fast and slow adaptation and reports the stability benefit of a two-timescale schedule. That bounds what a cadence claim can be worth, and we treat cadence as a control variable rather than a contribution.

Clustering trajectories. MEDS reuses layer-wise logits from the forward pass as lightweight representations of reasoning behaviour and clusters error patterns with HDBSCAN. We adopt this directly as the learned instantiation of our cluster key, against a derived key computed from the silence side of (2); the two are a controlled ablation of *how* to cluster, holding what is done with the clusters fixed. *Scope, narrowed by measurement.* A four-partition probe on exact per-cluster gradients finds MEDS clusters indistinguishable from a size-matched random partition, with the sign of the separation flipping across settings. We therefore do not claim that MEDS separates conflicting updates, and we do not build per-cluster experts on it. The use that survives is as a targeting key for *where* a supplier should act, which requires only that clusters identify consistent error patterns rather than separable gradients, and which is a weaker and separately testable condition. A confirmatory run at a later checkpoint is in flight.

Policies as weights and as code. Contextual bandits are the standard tool for per-instance algorithm selection under a scalar reward, and we use LinUCB unmodified; the contribution is not the estimator but what it is given to see. Generating a policy as executable source is the other established line, and we adopt it as a distinct arm rather than an alternative implementation, because the two differ in hypothesis class rather than in optimiser: weights reweight the features they are handed, code composes them.

Evaluation practice. Recent evaluation work documents how often agent results fail to survive controls that were available at the time. We adopt its checks as requirements rather than as discussion: paired tests on shared items, intervals reported on the *difference* rather than on each arm, matched-permutation controls that hold the proportion of units in each mode fixed while shuffling which units get which mode, and a noise floor established from the same checkpoint scored twice. The matched control is the one that decides our central claim, because a routing method changes both which units are routed and how many, and only the permutation control separates those.

3 Results

All numbers below are measured end-to-end on held-out benchmarks. Train reward is never reported as a result: Sec. 3.1 is the demonstration that it mis-orders checkpoints.

3.1 Train reward mis-orders checkpoints

Table 1 scores two recipes that differ only in optimiser aggressiveness, on the full MATH-500, paired by item. The aggressive (“demo”) recipe’s training reward rises while its held-out accuracy falls by 0.194 absolute. Any claim validated on training reward would have selected the collapsed checkpoint.

Table 1: MATH-500, $n = 500$, paired McNemar against the scaffold arm. Runs **step0d** (demo, lr 6×10^{-6} , clip 0.4) and **step0h** (scaffold, lr 1×10^{-6} , clip 0.2).

step	demo	scaffold	Δ	McNemar p
base	0.528	0.506	-0.022	same model
28	0.454	0.530	+0.076	7.3×10^{-4}
57	0.440	0.530	+0.090	1.7×10^{-4}
86	0.466	0.526	+0.060	4.1×10^{-3}
115	0.364	0.536	+0.172	2.1×10^{-13}
144	0.334	0.522	+0.188	9.3×10^{-15}

Table 2: The same checkpoints on OlympiadBench (675 problems), which shares no items with MATH-500.

checkpoint	MATH-500	OlympiadBench
base	0.528	0.188
gs028	0.454	0.170
gs057	0.440	0.190
gs086	0.466	0.148
gs115	0.364	0.135
gs144	0.334	0.108
gs173	0.358	0.107

Noise floor, and what it disqualifies. The base model appears in both series, so its difference is pure measurement noise: 0.0220 at $n = 500$. It *rose* from 0.0080 at $n = 250$ because the larger series was scored in two passes on different server processes, accumulating between-run variance a contiguous sweep does not have. Consequently the +0.060 at step 86 is only $2.7\times$ the floor and we do not report it standalone; the +0.172 and +0.188 are $\approx 8\times$ it. The claim is *preservation*, not improvement: the scaffold arm stays within [0.506, 0.536], which brackets its own base.

3.2 The collapse replicates on an independent benchmark

$0.188 \rightarrow 0.107$ is a 43% relative drop in the same direction as MATH-500’s $0.528 \rightarrow 0.358$, on an independent problem set.

Benchmark selection, stated as a measurement rather than a preference. MATH-500 is saturated at the frontier tier (0.966 at 27B) and AIME is unusable at both ends (0.000 at 1.5B; 0.867 at 27B with a 24-point interval on 30 problems). OlympiadBench resolves a few points on 675 items and leaves both ends off the rails: 0.188 at 1.5B and 0.716 for Ornith-1.5-35B-A3B. The 0.716 is a *lower* bound — 174/675 generations (26%) hit the token cap — so the headroom is smaller than the 28 points it suggests, and we quote it as a bound.

3.3 Negative result: token-level routing at 1.7% reach

Token-level routing changes nothing detectable. We report this rather than dropping it, because it is the measurement that located the method’s leverage. The rule reaches at most 1.7% of loss-carrying tokens, and an intervention on 1.7% of the gradient cannot move a benchmark by more than noise; the between-arm noise floor here (0.030 on the shared base checkpoint) is itself larger than every routing difference in the table.

Table 3: MATH-500, $n = 500$. **step0j** adds token-level routing to the scaffold recipe. p is paired McNemar, scaffold vs. scaffold+routing.

step	demo	scaffold	+routing	p
base	0.528	0.506	0.536	0.092
28	0.454	0.530	0.516	0.534
57	0.440	0.530	0.526	0.920
86	0.466	0.526	0.522	0.923
115	0.364	0.536	0.496	0.065
144	0.334	0.522	0.526	0.922

Table 4: Decomposition of the RL-silent channel. **step01**, GSM8K, Qwen2.5-1.5B-Instruct, first 18 logged batches.

quantity	mean	range
silent groups	0.3592	[0.2891, 0.4414]
silent because <i>solved</i>	0.3145	[0.2461, 0.4219]
silent because <i>unsolved</i>	0.0447	[0.0117, 0.0703]
solved share of the silent channel	0.875	[0.827, 0.959]

Confound, stated rather than buried. **step0j** differs from **step0h** in three ways — routing, `kl_ctl` ($0.01 \rightarrow 0$) and `adv_norm` (batch $\rightarrow$ off) — because the rule’s precondition does not hold otherwise. A significant difference here could not have been attributed to routing alone. The routing-off arm at **step0j**’s exact configuration is therefore a required control, and it is running (**step01**).

3.4 Where the leverage actually is

Measured live on **step01** at 64 groups per batch:

$$\hat{P}(\text{silent}) = 0.574, \quad n_{\text{groups}} = 64.$$

More than half of every batch contributes exactly zero gradient — $34\times$ the token-level channel, and the reason Sec. 3.3 was null is that the intervention was on the wrong tier, not that the idea was wrong.

A metric bug found and fixed, reported because the affected runs are still cited. The accompanying split first reported 0.000 solved at a measured 82% solve rate, which is impossible. It thresholded `reward_score`, which by that point has had bias, scaling, clipping and normalisation applied. It now reads the raw rewards captured before any transformation. Runs launched before the fix carry the old code and their solved/unsolved split must be ignored; $\hat{P}(\text{silent})$ is unaffected because it is computed from the advantages being zero.

3.5 The silent channel is 87.5% solved, so most of it needs no teacher

Measured on the routing-off control (**step01**) under the corrected metric, over 18 logged batches at 64 groups each (1152 group observations):

Table 5: Rollout cost of discarding versus reusing the silent channel. s is measured on the routing-off arm; the multiplier is $1/(1 - s)$.

point in run	silent fraction s	DAPO	ours
first 10 batches	0.327	1.49×	1.00×
whole run	0.525	2.10×	1.00×
last 10 batches	0.583	2.40×	1.00×
max observed	0.633	2.72×	1.00×

The identity solved + unsolved = silent holds to a maximum residual of 1.0×10^{-5} across all 18 batches, which is float32 rounding. We check it because an earlier version of this metric failed it, reporting 0.000 solved at a measured 82% solve rate.

This inverts the cost of the method. By (3), the solved groups already contain a supervised target, so 31.4% of *all* groups are reusable with no teacher, no extra inference and no extra GPU. Only 4.5% of groups — the unsolved ones — require an external teacher, and those are exactly the units for which the harness is the only available consumer. The reach of the free path is therefore about 7× that of the teacher-dependent one, which is the opposite of the ordering we assumed before measuring.

Scope, and what this does not claim. This is one operating point: GSM8K, Qwen2.5-1.5B-Instruct, early training, high solve rate. A harder task moves mass from solved to unsolved and shifts the balance toward needing a teacher; 0.875 is a property of the operating point, not a constant. More importantly, the table says where the reachable mass *is*, not that training on it helps. Sharpening an already-correct distribution spends entropy, and entropy collapse is the failure mode Sec. 3.1 already documents. Until the A/B exists, the solved branch of (5) defaults to SKIP.

3.6 What discarding the silent channel costs

DAPO’s response to the same channel is to drop it and regenerate. If a fraction s of groups is unanimous, refilling a batch requires generating $1/(1 - s)$ groups per accepted group. Substituting the silent fractions measured on the control arm (98 batches) gives Table 5.

The gap widens with training. The silent fraction is not a constant: it rises from 0.33 to 0.58 within 98 batches. Crucially this was measured on the arm with *no routing at all*, so the growth is a property of RL training rather than of our intervention — as the policy improves, more groups become unanimous and GRPO’s effective batch shrinks. An overhead that grows as the model gets better is the wrong shape for the regime frontier models occupy.

Stated as a prediction, not a result. $1/(1 - s)$ is the expectation under the assumption that regenerated groups are unanimous at the same rate. The DAPO arm measures the true multiplier from the accept/reject counts rather than inheriting this estimate. We fix the comparison axis in advance as *matched generation budget*, not matched steps: DAPO’s kept batch is denser per step, so an equal-step comparison would flatter it.

Table 6: The silent channel does not shrink on a harder task — it inverts. Qwen2.5-1.5B-Instruct in both columns.

	GSM8K	MATH
silent groups	0.359	0.419
silent because solved	0.315	0.164
silent because unsolved	0.045	0.255
solved share of the channel	87.5%	39.1%
unsolved share	12.5%	60.9%

3.7 The solved branch is inert, and then harmful

Acting on the free self-target does not help. At $c_+ = 0.5$ the effect is null at every checkpoint and the difference-in-differences after removing the base offset is 0.000–0.002 at four of five steps. At $c_+ = 2.0$ it becomes negative: -0.054 at step 144 ($p = 0.014$, $2.5\times$ the noise floor). We report it as suggestive rather than conclusive — one hit in five paired tests does not survive a Bonferroni threshold of 0.01 — but the *shape* argues it is real: negative or zero at four of five steps with the largest effect at the *last* checkpoint, which is what cumulative over-sharpening looks like and is not what a random hit looks like. It agrees with an independently measured entropy trace in which the routed arm descends faster mid-run.

The mechanism is unflattering and simple: a group is solved *because* the policy already places high probability on the correct answers, so supervising on them teaches what the model already knows.

3.8 The composition of the silent channel inverts with task difficulty

If that were the whole story the channel would be worthless, because on GSM8K it is 87.5% solved. It is not the whole story. Holding the model and configuration fixed and changing only the training task:

The unsolved fraction of *all* groups rises from 4.5% to 25.5%, a $5.7\times$ increase, while the solved fraction roughly halves. So the branch we measured to be worthless shrinks to a minority exactly where the task gets hard, and the branch for which no self-target exists — and for which a teacher or the harness is the only possible consumer — becomes the majority.

This is what the routing rule of (5) is for. A fixed threshold tuned on GSM8K would spend its effort on the solved side, which is where the effort is wasted; keying on the *side* of silence rather than on a tuned constant is what survives the shift. We state the limits plainly: $n = 11$ batches for the MATH column against 98 for GSM8K, one model at one scale, and — the lesson of Sec. 3.7 — locating mass is not the same as showing that acting on it helps.

3.9 The learned router does not beat its matched control

Sec. 1.5 names the control that decides this claim: `router=random` run at the contextual arm’s *measured* mode proportions, so the two arms differ only in *which* unit receives which mode and not in the mixture. Those proportions were read off the last 40 steps of the contextual run (`ctx`, 64 groups per step) — `rl` 0.2946, `sft` 0.3527, `skip` 0.3526 — and the control (`rnd`) was run at exactly them. Both arms are scored from `globalstep149` of their own run, GSM8K training, Qwen2.5-1.5B-Instruct, greedy decoding (temperature 0, seed 0). That step is the matched comparison point rather than a

Table 7: Learned router (**ctx**) against its matched-proportion control (**rnd**), both at **globalstep149**, greedy. Wilson 95% intervals in brackets. MATH-500, AMC23 and AIME are scored at **max.tokens=3072** and OlympiadBench at 16384; arms are compared only against each other at the same budget.

benchmark	n	ctx	rnd	ctx – rnd
MATH-500	500	0.5240 [0.480, 0.567]	0.5260 [0.482, 0.569]	−0.0020
OlympiadBench	675	0.1837 [0.156, 0.215]	0.1837 [0.156, 0.215]	+0.0000
AMC23	40	0.1750 [0.087, 0.320]	0.1250 [0.055, 0.261]	+0.0500
AIME24	30	0.0000 [0.000, 0.114]	0.0000 [0.000, 0.114]	0.0000
AIME25	30	0.0000 [0.000, 0.114]	0.0333 [0.006, 0.167]	−0.0333

chosen one: the contextual run stalled at step 162 of 290 while the control reached 245, so 149 is the latest checkpoint the two arms share.

Stated positively rather than as an absence: **at matched mode proportions, the per-unit routing decision buys nothing over assigning the same modes at random.** What this routing arm sets is the *mixture*; which unit receives which mode carries no effect the suite can measure.

Which rows can carry that claim, and which cannot. Only the first two. MATH-500’s −0.0020 is one tenth of the 0.020 systematic noise floor measured for greedy scoring on this suite (Sec. 3.1 measures 0.0220 on MATH-500 itself), and OlympiadBench returns the same score on both arms to four decimals over 675 problems. The other three rows resolve nothing: at $n = 40$ and $n = 30$ the Wilson intervals span more than 20 points, so AMC23’s +0.0500 and AIME25’s −0.0333 — one problem out of thirty — sit well inside them, and AIME24’s 0.000 on both arms is this scale’s known floor rather than a comparison. One caveat on the intervals themselves: they are per-arm Wilson intervals, not the paired intervals on differences this section reports elsewhere. The paired test was not run for this pair, so what bounds the two resolving rows is the noise floor, not an interval on the difference.

What a real effect on OlympiadBench would have to clear. A single greedy score at $n = 675$ carries about a point of jitter: the *same ctx* checkpoint on the *same* 675 problems scored 0.1941 at **max.tokens=8192** and 0.1837 at 16384, with nothing changed but the budget. A future arm difference below roughly two points on this benchmark is therefore not evidence, and +0.0000 is far inside that. Both arms also leave generations that never emit a boxed final answer at any budget — 78/675 for ctx and 64/675 for rnd at 16384, against 79 and 61 at 8192 — so the truncation these rows carry is non-termination rather than a budget-bound score, and doubling the cap does not remove it.

Scope, and what this does not claim. One seed, one checkpoint, one training task, one model scale. It falsifies “this router, with this credit signal, beats its matched control”. It does not show that routing cannot help, and by construction it says nothing about the mixture, on which the two arms are matched. Sec. 3.10 localises the failure, and the failure is not the bandit.

3.10 Why the router does not learn: the credit signal, not the bandit

This subsection and the next report the controller’s own training-time behaviour — the mode mix it emits — and not capability. They are diagnostics of a mechanism, not results on a held-out

Table 8: Mode mix under per-batch credit. `ctx`, GSM8K, Qwen2.5-1.5B-Instruct, first 129 logged steps at 64 groups per step. L_1 is the distance of the quarter’s mean mix from uniform thirds; 1.33 would be total concentration on one mode.

quarter (steps)	rl	sft	skip	L_1 vs. uniform
Q1 (1–32)	0.305	0.341	0.354	0.056
Q2 (33–64)	0.299	0.338	0.363	0.069
Q3 (65–96)	0.351	0.303	0.346	0.061
Q4 (97–129)	0.344	0.320	0.336	0.027

benchmark, and are labelled as such throughout.

Under per-batch credit the router never developed a preference. Table 8 gives its mode mix by quarter over the first 129 logged steps of `ctx`.

The mix stays at uniform thirds and the trend is flat to *decreasing* — the last quarter is the most uniform of the four. A learner should move *away* from uniform as its arm estimates separate; this does the opposite.

Feedback was flowing, not blocked. The absence is not a plumbing failure. Over those 129 steps the router received 128 feedback records, each covering all 64 units of its batch, with exactly one batch in the whole run refused as confounded (a batch whose units all took one mode carries no comparison, and its update is skipped rather than made). The batches were not degenerate either: the most common mode’s share stays between 0.352 and 0.766; all three modes are present in 95 of the 128 records, and the count never falls below 2.25; and the weak-attribution flag, which fires when one mode exceeds 90% of the units, reads 0 in 108 of the 128 and never exceeds 0.5. Both counts are averaged over four data-parallel ranks, which is why they are fractional: a value below 3 means one rank’s view of the batch was short a mode, not that the batch itself was. The router got 128 updates on batches that carried a comparison and formed no preference from them.

The mechanism, as an argument. LinUCB accumulates, for the arm m that was chosen,

$$A_m \leftarrow A_m + xx^\top, \quad b_m \leftarrow b_m + r x, \quad \theta_m = A_m^{-1} b_m.$$

Per-batch credit gives every unit in the batch the same r , and under a cold start that assigns modes independently of the context x , an arm chosen n_m times accumulates $A_m \approx n_m \mathbb{E}[xx^\top]$ and $b_m \approx r n_m \bar{x}$, so

$$\theta_m \approx (n_m \mathbb{E}[xx^\top])^{-1} r n_m \bar{x} = \mathbb{E}[xx^\top]^{-1} r \bar{x},$$

which is *the same vector for every arm*: the per-arm count cancels. The arms start identical and stay near-identical, selection is left to the exploration bonus, and the mix stays uniform. This rules out “the router needs more steps”: an uninformative signal does not become informative with repetition.

The mechanism, as a measurement. The argument is turned into a test in `selfevo/tests/test_credit_assign` which drives the same registry-built router for 60 batches of 64 units and changes *only* the credit (seed 0, the default the test runs at). Under a shared per-batch scalar the largest pairwise distance between two arms’ parameter vectors is 0.124 and no mode takes more than 0.379 of the units (the test asserts < 0.5 and < 0.55). Under credit that depends on which mode was applied, the same router separates to 1.500 and the rewarded mode takes 0.989 of the units (asserted > 1.0 and > 0.5).

Table 9: Mode mix under per-prompt credit. `ctxpc`, GSM8K, Qwen2.5-1.5B-Instruct, first 85 logged steps at 64 groups per step; the run was still training when these were read. Compare L_1 against 0.027–0.069 for per-batch credit (Table 8).

quarter (steps)	rl	sft	skip	L_1 vs. uniform
Q1 (1–21)	0.905	0.046	0.049	1.144
Q2 (22–42)	0.381	0.405	0.214	0.239
Q3 (43–63)	0.000	0.353	0.647	0.667
Q4 (64–85)	0.006	0.431	0.562	0.655

Table 10: RL share of the batch under raw (`ctxpc`) and batch-mean-centred (`ctxpcc`) per-prompt credit, matched over the first 89 logged steps of each arm.

steps	raw	centred
1–15	0.867	0.870
16–30	0.933	0.933
31–89	0.003	0.001

Same bandit, same contexts, same number of updates, opposite outcome — so the null of Sec. 3.9 localises to the credit signal, not to the controller or its exploration.

3.11 Per-prompt credit changes the router’s behaviour; centring the credit does not

The implied fix is to credit each decision with its own prompt’s change in solve rate between the batch where it was routed and that prompt’s next appearance, rather than with one scalar per batch. Table 9 is the same measurement as Table 8 on the arm that does this (`ctxpc`, identical to `ctx` except for the credit).

The mix stops being uniform, and it revises: L_1 swings between 0.239 and 1.144 where per-batch credit never left 0.027–0.069. Only the credit changed — same router, same features, same exploration — so the difference is attributable to the signal.

The transition is abrupt, and it coincides with the first credit. Per-prompt credit reaches nothing until a prompt recurs. `prompt_credit/observed_units` is exactly 0 for steps 1–28 and first non-zero at step 29 (4.25 of 64 units); the first step whose `rl_groups` is 0 is step 30, the step immediately after. Before the credit arrives the arm is not routing on evidence at all: after three cold-start steps near uniform thirds (0.352, 0.328, 0.332 rl), steps 4–29 sit at exactly 1.000 rl. That is the documented tie-break artifact — with no `observe` calls the arm parameters never move, the scores tie, and `argmax` resolves alphabetically to RL — so the pre-transition plateau is not a preference either.

Centring the credit changes essentially nothing, which kills the obvious confound. The credited value is a prompt’s change in solve rate across roughly 29 steps of training, over which the policy improves generally; the signal therefore conflates “the model got better” with “this mode helped this prompt”. Subtracting the batch’s mean delta removes the common component. Matched over the first 89 logged steps of each arm:

The two arms are the same arm to within a third of a point in every window, and the common trend that centring removes is small: `prompt_credit/mean_delta` is exactly 0 for the 28 steps before

Table 11: One 30B checkpoint scored at the cap that was requested and at the cap that was applied. **cap asked** is what the command line requested — the same value in every row — and **cap used** is what the sampler received. Nothing else changed.

benchmark	n	cap asked	cap used	accuracy	truncated
AIME24	30	32768	8192	0.2667	73.3%
AIME24	30	32768	32768	0.8000	20.0%
AIME25	30	32768	8192	0.2667	73.3%
AIME25	30	32768	32768	0.7000	20.0%

credit flows and lies in 0.005–0.090 (median 0.049) afterwards. Removing the common trend does not move the behaviour, so **the training-progress confound is not the explanation** for the collapse away from RL.

The surviving hypothesis, labelled as one. What remains is that the transition is driven by the *exploration bonus* rather than by the credit values: the heavily-used arm is the first to have its A updated, so its confidence bonus is the first to shrink, and the untouched arms become relatively more attractive whatever the credit says. This predicts exactly the value-independence observed above, and it predicts self-correction once the bonuses equilibrate. The second prediction is at least not contradicted — the raw arm’s rl share returns from 0.000 across steps 43–63 to a mean of 0.336 over steps 120–132 (0.26 at step 130) and 0.367 over steps 115–152, the last step logged when this was read — but that is an observation on a run that is still training, not a test. **This has not been tested.** Two things would settle it. Logging the two terms of the LinUCB score separately at the transition: bonus exhaustion requires that the point estimates $x^\top \theta_m$ are still tied when the ranking flips, and is dead if they had already separated in favour of SFT and SKIP before step 30. And re-running the same credit with the exploration coefficient at zero: if the transition survives, it was never the bonus.

Scope, and what this does not claim. One arm per credit variant, one seed, one task, one scale, and both runs unfinished at the step counts quoted. Nothing here is a capability claim: **neither credit arm has been scored on the held-out suite**, and a mix that moves is not a mix that helps — especially this one, which drives **rl_groups** to zero, so that no group receives an ordinary policy gradient and every group is either supervised on its own correct sample or skipped. Sec. 3.7 measures that branch inert at $c_+ = 0.5$ and harmful at $c_+ = 2.0$, which is a reason to expect this regime to cost capability rather than to gain it. Until a scored arm exists, this is a demonstration that the credit signal controls the router’s behaviour, and nothing more.

3.12 A benchmark score is a property of the token budget it was taken at

The scoring harness resolved generation parameters by applying a per-benchmark defaults table unconditionally, so an explicit **--max-tokens** was parsed, accepted, and then discarded. A run made with the wrong budget still succeeds and still writes a plausible number, so the defect is invisible from the score alone. Table 11 re-scores one 30B checkpoint (Frontis-MA1-30B, a Qwen3MoeForCausalLM mixture of experts) after fixing the precedence: same model, same server, same concurrency, same grader, and the only thing that differs between the two halves of each block is which cap reached the sampler.

Table 12: Length-knee onset across six GSM8K runs on Qwen2.5-1.5B-Instruct. c_+ is the constant written onto solved groups. “steps” is the number of logged training steps available for the run, so the control rows record how long a crossing had to fail to appear.

run	group routing	c_+	steps	onset
ctx	on	0.5	162	144
ctxpc	on	0.5	152	132
sa2	on	2.0	145	131
step0l	off	—	273	none
step0j	off	—	178	none
g16	off	—	116	none

AIME24 moves by a factor of three and AIME25 by a factor of 2.6, from a flag that appeared on the command line of every earlier score and never reached the sampler.

Both corrected rows are still lower bounds. 20% of generations reach even the 32768 cap and are graded wrong, so 0.8000 and 0.7000 bound this checkpoint from below rather than measuring it. OlympiadBench carries a second and unrelated lower-bound caveat at every cap: 94 of its 675 problems state more than one gold answer inside a single answer string, which the exact-match grader marks wrong however the model answers.

What made the defect visible, and it was not the score. The corrected code prints the budget it resolved (NOTE ... BENCH_OVERRIDES default 8192 not applied), and the results row records the generation parameters the run actually used, so the requested cap can be compared against the applied one. Recording the parameters a run used is not bookkeeping; without it a cap defect produces no symptom at all, because the run still completes and the number it writes is still in range.

The general form, which outlives this particular bug. A default that silently outranks an explicit request is indistinguishable, from outside, from the request being honoured. We therefore state the convention rather than assume it: **a benchmark number is a property of the token budget unless the budget is reported**, and a comparison taken across two budgets is not a comparison.

3.13 Length moves like a threshold, and routing *presence* is what separates

Mean response length during training does not drift upward under routing; it holds a plateau and then breaks out of it. We fix the onset criterion before reading the runs: onset is the first step of five consecutive steps whose mean response length exceeds the run’s own pre-onset baseline by more than four of that baseline’s standard deviations.

All three runs with group routing configured hold a flat plateau for more than 120 steps and then cross; none of the three runs without it ever crosses, over 273, 178 and 116 steps. Slope brackets the change: 0.34–0.57 tokens per step before onset across the three routed runs, and 5.6–9.2 after.

The same separation appears in free generation. Scored outside training, greedily, from the checkpoint ladders: ctx grows 327 → 455 tokens, while step0l moves 330 → 316 and is flat over 202 steps.

Presence, not magnitude. We state the surviving claim in the narrowest form the data supports. Whether group routing is *configured* predicts whether a run crosses; the *magnitude* of the constant does not predict when it crosses, which is the negative result of Sec. 3.14. Three runs on each side is enough to notice a clean split and not enough to estimate an effect, and the six runs are not a matched sweep: `g16` trains at $G = 16$ where the others use $G = 8$, `step0j` carries the token-level rule of Sec. 3.3 and is unrouted only at the group level, and the runs were allowed different numbers of steps.

How each run was classified, because the obvious way is wrong. Routing status is read from `actor.group.routing` in each run’s `config.yaml`, never from the presence of `route/*` metrics in its log. `sa2` emits no `route/*` metric at all while its config carries `group.routing.enabled: true` with `solved.advantage: 2.0` and `router: null` — the actor’s fixed-rule branch writes the constant without passing through the metric seam. Classified from metrics alone, `sa2` would have been counted on the wrong side of Table 12, and it is the run carrying the larger constant.

3.14 Three negative results on the routed arms

(a) **The learned router shows no preference over its matched random control.** Sec. 3.9 is that result and we do not restate its table: at the contextual arm’s own measured mode proportions the per-unit decision is worth -0.0020 on MATH-500 and $+0.0000$ on OlympiadBench, against a 0.020 noise floor, and Sec. 3.10 localises the cause to the credit signal rather than to the controller.

(b) **The dose-response between constant magnitude and knee onset is not supported.** We had recorded — and described as the strongest evidence available — that the larger the constant written onto solved groups, the earlier the length knee arrives. Under the onset criterion of Sec. 3.13 it does not hold. $c_+ = 0.5$ gives onsets at steps 132 (`ctxpc`) and 144 (`ctx`); $c_+ = 2.0$ gives 131 (`sa2`), which sits inside the spread of the two 0.5 arms rather than ahead of them. Three runs at two operating points cannot resolve a dose-response, and the earlier claim is withdrawn here rather than quietly not repeated.

(c) **The runs that genuinely produced non-terminating output were the *unrouted* ones.** Reading training-time `no_eos_ratios` across all 18 training runs logged for this study, the routed arm `ctx` peaks at 0.0059 — six of 1024 rollouts. The two runs that produced non-terminating output at any material rate are `step0b` (0.156) and `step0` (0.117), and **both have group routing off**. Both use the same aggressive hyperparameters as the demo recipe of Sec. 3.1 ($\text{lr } 6 \times 10^{-6}$, clip 0.4), which that section already measures to destroy capability.

The caveat that keeps (c) from being stronger than it is. `no_eos_ratios` is mis-instrumented: it compares sequence lengths against the *padded* batch width, so it can only count rows that reach that width and it under-counts. Every value it reports is a lower bound. That is enough to establish the positive half of (c) — `step0b` and `step0` really did produce non-terminating output — and not enough on its own to establish the negative half. What supports the negative half is Sec. 3.15, which measures the routed arm’s stop probability directly, and the routed arm’s free greedy output, which grows to 455 tokens against a 1024 training cap rather than running away from it.

Table 13: Integrated EOS hazard over 64 frozen greedy responses, teacher-forced through the routed ladder (**ctx**) and the group-routing-off control (**step01**). Four of the 27 probed checkpoints, at steps both ladders share.

step	ctx (routed)			step01 (control)	
	H	p_{end}	H_{body}	H	H_{body}
24	0.99754	0.99754	$< 5 \times 10^{-7}$	0.99924	$< 5 \times 10^{-7}$
74	0.99958	0.99958	$< 5 \times 10^{-7}$	1.00204	0.00208
99	0.99975	0.99976	$< 5 \times 10^{-7}$	1.20534	0.20538
149	0.99958	0.99958	$< 5 \times 10^{-7}$	1.09105	0.09110

3.15 Teacher-forced EOS hazard: the routed arm’s termination signal does not erode

Sec. 3.13 leaves two candidate mechanisms that the training logs cannot separate, and both are statements about the probability of stopping. Under *smooth erosion*, an objective that raises the log-probability of tokens that were produced gradually lowers the probability the model assigns to the stop token. Under a *dynamical threshold*, that probability holds and then collapses. We measured the quantity rather than argued about it.

Protocol. One set of 64 greedy responses is generated once and frozen. Each of 27 checkpoints — the routed ladder (**ctx**), the matched control with group routing off (**step01**), and the base model as step 0 — is then teacher-forced over those same token sequences, and we read the model’s probability of the stop token at every position. Writing T for the position of the true final token,

$$H = \sum_{t \leq T} p_{\text{eos}}(t), \quad p_{\text{end}} = p_{\text{eos}}(T), \quad H_{\text{body}} = H - p_{\text{end}},$$

so H is the integrated stop hazard over a response, p_{end} is the spike at its semantic end, and H_{body} is stop-mass placed anywhere else.

Two things pinned before the numbers mean anything. The stop token was determined empirically rather than assumed: all 64 frozen responses end on 151645 (`<|im_end|>`) and none on 151643. Alignment between the scored positions and the text was checked by the strongest test available here — the frozen responses are greedy, so a correctly aligned teacher-forced pass must reproduce them by argmax. Agreement at the generating checkpoint is 0.998, with $p_{\text{end}} = 0.99928$; an off-by-one gives approximately zero on both.

The routed arm’s hazard does not erode. Across the ladder it rises, from 0.99754 at the earliest probed checkpoint to a maximum of 0.99979, and the minimum over the 64 prompts rises with it, $0.850 \rightarrow 0.994$. H_{body} stays below 5×10^{-7} at every probed step, so the hazard is a spike at the response’s semantic end and that spike does not move. We quote the range rather than a trend because the four steps in Table 13 are not themselves ordered (0.99975 at step 99 against 0.99958 at 149, a difference of 1.7×10^{-4}); what the ladder supports is that H stays within 2.5×10^{-3} of 1 throughout and at no point declines toward zero.

This refutes both mechanisms we had entertained. Smooth erosion predicts a falling H , and it does not occur. The dynamical-threshold account is refuted at its premise rather than at its prediction: the hazard is not the failing state variable at all, because it never leaves a 2.5×10^{-3}

band while the same arm’s free greedy output grows from 327 to 455 tokens. Whatever the length increase is, it is not the model losing its ability to stop.

Not a stale-text artifact. The responses are frozen, so a sufficiently drifted checkpoint could in principle be scored on text it would never itself produce. A self-generated control repeats the measurement with each checkpoint scored on its *own* greedy output ($n = 32$): `ctx`’s mean p_{end} stays in $[0.984, 0.9999]$ while its own greedy length grows $324 \rightarrow 469$.

The control is not flat, and we had claimed it was. `step01`’s H rises to 1.205 at step 99. Its end spike is pinned at 0.99996 throughout, so the extra 0.205 of stop-mass sits *mid-response*, on 16 of the 64 responses, reaching 1.81 on one. The control drifts toward stopping *earlier*, which is the opposite direction from the routed arm’s length growth and therefore does not confound the comparison — but the claim that it was flat was wrong, and is corrected here rather than dropped.

What this retires on the fix side. The codebase offers exactly two explicit termination terms, `overlong_reward_penalty` and `mask_no_eos_with_zero`, and both are disabled in every arm reported in this paper. Either one acts on p_{eos} , which is already 0.9996 at the true end and does not move. There is nothing there for it to correct, so if the routed constant needs to be made safe the lever is the constant itself, not a termination penalty added beside it.

Limitations of the termination and length measurements. Four, stated together because each bounds Secs. 3.13–3.15. (i) *The post-knee regime is unmeasured.* `ctx`’s last saved checkpoint is step 149; the run continued to 162 with mean length still climbing (647 and rising) and was then killed abruptly rather than stopped on a criterion, so a hazard collapse living past the last checkpoint is not something these probes could see. (ii) *The probes are greedy and the training was not.* The frozen and self-generated sets are both greedy, while training sampled at temperature 1.0. Greedy length at step 149 (455) tracks the training-time sequence length at step 145 (478), so greedy is a fair proxy for the mean, but the sampled tail is not measured. (iii) $n = 64$ prompts for the frozen probe and 32 for the self-generated control: the hazard differences we are calling null are small, and so is the sample they are null over. (iv) *The control is itself moving*, as the paragraph above records, so Table 13 compares a flat routed arm against a drifting control rather than against a fixed reference.

3.16 Frontier comparison at an honest cap

Sec. 3.12 establishes that a benchmark number without its budget is not a measurement. Table 14 applies that convention to the two frontier candidates for the fixed strong base: `Qwen3.8-27B` and `Frontis-MA1-30B`, both scored on the same box with the same grader at an explicit 32768 cap, with the cap-precedence fix of Sec. 3.12 confirmed live on that box by the line the corrected code prints, `NOTE aime24: using explicit --max-tokens=32768` followed by `(BENCH_OVERRIDES default 8192 not applied)`. Truncation is reported beside every accuracy, because at this cap it is the column that says whether the accuracy beside it is a measurement or a lower bound.

The truncation column carries as much of the comparison as the accuracy column. `Qwen3.8-27B` is 0.8% cap-limited on MATH-500 against `Frontis-MA1-30B`’s 11.4%, so its MATH-500 figure is close to a measurement while every `Frontis-MA1-30B` figure in the table remains a lower bound. The smaller model wins on all five benchmarks and is less cap-limited on all five,

Table 14: Frontier comparison at a matched, explicit 32768 cap, greedy, one run per model. **trunc** is the number of generations that reached the cap and were therefore graded wrong. AMC23 is included for completeness and is read as saturated rather than as a score.

benchmark	n	Qwen3.8-27B		Frontis-MA1-30B	
		accuracy	trunc	accuracy	trunc
MATH-500	500	0.9760	4/500 (0.8%)	0.8980	57/500 (11.4%)
AMC23	40	1.0000	0/40	0.9500	2/40
AIME24	30	0.9000	4/30 (13.3%)	0.8000	6/30 (20.0%)
AIME25	30	0.9000	3/30 (10.0%)	0.7000	6/30 (20.0%)
OlympiadBench	675	0.7733	120/675 (17.8%)	0.7363	135/675 (20.0%)

which is why the paper’s fixed strong base is Qwen3.8-27B: a delta measured on it is both harder to obtain and harder to dismiss as scale.

AMC23 at 1.0000 is saturation, not a score. With $n = 40$ and zero errors the Wilson interval is $[0.912, 1.000]$, so the benchmark cannot separate two models that both sit here and the row resolves nothing. We report it and then decline to use it: AMC23 has no resolving power left at this capability and is excluded from any comparison that has to separate strong models.

A matched cap is not automatically a fair cap. Both models truncate at 32768, so neither number measures capability; both measure *capability under a 32768 budget*, which is a legitimate quantity but not the one a bare ranking implies. Where a cap binds on both sides, part of the gap is verbosity rather than skill, and the ranking is trustworthy only if it is stable as the cap grows. On OlympiadBench Frontis-MA1-30B is visibly still on the slope — 0.6237 at 16384 (33.3% truncated), 0.7363 at 32768 (20.0%), with 65536 not run — so that row in particular is a comparison of two lower bounds. Sec. 3.17 supplies the stability check for the other four.

Scope. One run per model, greedy, one set of caps. AIME at $n = 30$ carries about 0.1 of run-to-run jitter on this pipeline, so 0.9000 against 0.8000 on AIME24 is about one jitter width and is not significant on its own; the MATH-500 gap, 0.9760 against 0.8980 at $n = 500$, is well outside it and is the row that carries the ranking. Both OlympiadBench figures inherit the grader caveat of Sec. 3.12: 94 of the 675 problems state more than one gold answer in a single answer string and are marked wrong however the model answers.

3.17 The 32768 cap is saturated on three benchmarks of four; OlympiadBench is the exception

The fairness objection to Table 14 is that a cap which binds on both sides partly measures verbosity. The check is to raise it. Table 15 re-scores Qwen3.8-27B at 65536: same model, same grader, same box, only the cap changed.

Correction to an earlier version of this subsection. The OlympiadBench row read *not yet measured* while its 65536 run was still going, and the subsection was titled as though saturation held on all four benchmarks. It does not. That row is now measured, it contradicts the old title for that benchmark, and the title and the framing below are the correction. The three saturated rows are unchanged.

Table 15: **Qwen3.8-27B** at 32768 against the same model at 65536, greedy, one run per cell. On MATH-500 and both AIME sets truncation more than halves and the accuracy does not follow, which is saturation. On OlympiadBench the accuracy does follow. Both OlympiadBench cells graded 675/675 with zero request failures.

benchmark	@32768	trunc	@65536	trunc	Δ
MATH-500	0.9760	0.8%	0.9800	0.2%	+0.004
AIME24	0.9000	13.3%	0.9333	6.7%	+0.033 (1 problem)
AIME25	0.9000	10.0%	0.9333	3.3%	+0.033 (1 problem)
OlympiadBench	0.7733	17.8%	0.8089	10.7%	+0.036

Three of the four are saturated at 32768. MATH-500 moves +0.004 at $n = 500$, where the standard error is about 0.006 — inside noise. Each AIME set moves by exactly one problem out of thirty, well inside the ≈ 0.1 run-to-run jitter recorded at that n . For those three the informative quantity is the pair rather than either number: truncation more than halves while accuracy does not follow, which is the signature of a budget that was already sufficient, and the generations being cut off at 32768 were ones that were going to be graded wrong regardless. Their 32768 figures in Table 14 are therefore measurements rather than lower bounds, and the ranking on those three rows can be stated plainly.

OlympiadBench is not saturated, and it is the largest cap effect of the four. Doubling the budget buys +0.036 there, with truncation falling 17.8% $\rightarrow$ 10.7%, on 675 problems graded with zero request failures on both rows. For this benchmark 32768 was not the plateau, and the generations it cut off were ones that would have been graded right. We attach no significance statement to +0.036: no repeat run of this benchmark at this cap exists, and the repeat spreads of Sec. 3.18 were measured on other checkpoints and another benchmark. What the row supports is the qualitative claim that here the accuracy tracks the truncation and elsewhere it does not.

The exception falls where a cap argument predicts it. Ordered by truncation at 32768 the four are MATH-500 0.8%, AIME25 10.0%, AIME24 13.3%, OlympiadBench 17.8%, and the only one that gains from a larger budget is the one whose truncation was highest to begin with. That is the saturation claim read in the other direction: a cap can only be costing accuracy where it is binding, so “the budget is sufficient” is a statement about a benchmark and not about a model, and it has to be checked per benchmark. 10.7% is still not zero, so the 65536 OlympiadBench figure is itself not fully saturated and remains a lower bound; Sec. 3.21 carries it into the comparison it was run for.

A caveat that belongs with these numbers. The 65536 MATH-500 and AIME runs used concurrency 40 against the earlier runs’ 24, so the batch composition differed between the two columns on those three rows. Decoding is greedy and the requests are independent, so no answer depends on which others ran beside it; but batched matmuls do not associate identically across batch sizes, and a one-problem AIME difference is exactly the resolution at which that could contribute. This does not weaken the conclusion, which rests on the accuracy *not* moving while truncation halved, and not on the sign of +0.033. The OlympiadBench row is the exception in the useful direction: after the client-timeout failure of Sec. 3.20 it was re-run at concurrency 24, the same concurrency as its 32768 row, so that pair is matched on batch composition as well as on cap.

Table 16: An accidental repeat run. Four checkpoints from the `ctx2` (contextual) and `rnd` (random-router control) ladders, each scored twice on MATH-500 at a matched 16384 cap. *spread* is $|A - B|$.

checkpoint	run A	run B	spread	truncation
<code>rnd@173</code>	0.4600	0.4580	0.002	10%
<code>ctx2@173</code>	0.4580	0.4480	0.010	11%
<code>ctx2@199</code>	0.2660	0.2420	0.024	96%
<code>rnd@199</code>	0.3480	0.3220	0.026	99%

Table 17: MATH-500 truncation at a matched 16384 cap across both ladders. Accuracies are deliberately not tabulated past the transition: by Sec. 3.18 they are not comparable there, and truncation is the trustworthy channel.

step	149	173	174	199	202	231	260	289
<code>ctx2</code> (contextual)	6.4%	11%	13.2%	96%	99.4%	92.2%	77.8%	96.4%
<code>rnd</code> (random)	6.0%	10%	12.4%	99%	99.6%	99.6%	97.4%	98.2%

3.18 The noise floor is a function of truncation

`narrow.sh` was launched twice by mistake, so four checkpoints were each scored twice on MATH-500 at a matched 16384 cap. The duplication measures the run-to-run spread of this exact pipeline directly, rather than by assertion, and it does so at two very different truncation rates.

At $\approx 10\%$ truncated the same checkpoint reproduces to 0.002–0.010; at $\approx 97\%$ truncated it moves by 0.024–0.026, an order of magnitude worse. The noise floor is therefore not one number for this suite — it is a function of how much of the batch the cap is cutting off.

The rule this licenses. Above roughly 80% truncation an accuracy is dominated by which few generations happened to finish, and finishing is not random: the survivors are the fastest generations, hence the shortest. **We do not compare accuracies taken above $\approx 80\%$ truncation**, and where two such points differ we read the truncation channel instead. This had until now been argued rather than measured; Table 16 is the measurement.

What it retires. We had recorded that the random-router arm *retains more accuracy* than the contextual one after the transition of Sec. 3.19 — at step 289, 0.3260 against 0.1740. Both points sit at 96–98% truncation, where the repeat spread is 0.025, so what that comparison reports is which few generations finished on each arm rather than a capability difference. The sign survives; the magnitude does not, and neither does any ordering among the post-transition points. We withdraw the reading here rather than quietly not repeating it.

3.19 The collapse transition narrows to steps 174–199, and both arms cross together

Every point in Table 17 is MATH-500 truncation at a matched 16384 cap, on the checkpoint ladders of the contextual arm (`ctx2`) and the random-router control (`rnd`). The cap is 16384 and not 32768 because these are 1.5B checkpoints with `max_position_embeddings: 32768`; asking such a model for 32768 *new* tokens is rejected outright (Sec. 3.20). It is $5.3\times$ the 3072 these arms were originally scored at.

The unmeasured window is twenty-five steps wide, and both arms cross inside it. Truncation goes $13.2 \rightarrow 96$ on the contextual arm and $12.4 \rightarrow 99$ on the random one between step 174 and step 199. The two ladders agree to within a percentage point at every measured step on either side of that window — 12.4 against 13.2 before, 99.6 against 99.4 after.

This is the strongest form of the controller-is-not-the-cause result. The random router has no learned policy, no credit signal and no preference over modes (Sec. 3.9), so the timing of this transition cannot be a property of the controller. It is a property of the routed constant, which both arms share. The earlier evidence for that conclusion was that both arms collapse; this is the stronger statement that they collapse *at the same step*.

It is also not a cap artifact, which was a live possibility. The original step-289 scores were taken at 3072 with $\approx 100\%$ truncation, where an accuracy is a truncation rate wearing accuracy’s clothing. Raising the budget $5.3\times$ moves truncation from $\approx 100\%$ only to 96–98%. These policies genuinely emit more than 16384 tokens; the pathology is in the model and not in the budget.

Two wrong shapes from one missing interval, recorded because the error repeats. This table has now corrected the reading of the same transition twice, in opposite directions. From training-time length curves on runs killed at step 162 we first called the change *threshold-like*; from four points at 149/174/260/289 we then corrected that to *gradual, not a cliff*; filling in 202, 231, 173 and 199 shows it is *sharp*. Both wrong shapes came from drawing a line through the same unmeasured gap. The general form is worth more than the particular correction: interpolating a shape across an interval that contains no measurement produces a confident claim whose sign is set by which endpoints happened to exist.

What the table does not support. There is no clean post-collapse plateau. After the transition the contextual arm reads $99.4 \rightarrow 92.2 \rightarrow 77.8 \rightarrow 96.4$, non-monotone across four points; its step-249 checkpoint (not shown, outside the matched step set) is the oddest, at 61.4% truncation with the ladder’s lowest accuracy, so at that checkpoint the failures are not all length. That is unexplained and we do not model it.

3.20 Three ways a harness reported a number that was not a measurement

Sec. 3.12 reports one defect of this kind. It was not the only one, and the three together have a shape worth stating as a contribution rather than as an apology: in all three the run completed, exited zero, and wrote a plausible number, so nothing downstream of the score could have caught them.

(i) **A defaults table silently outranking an explicit cap.** `resolve_params` applied a per-benchmark `BENCH_OVERRIDES` table unconditionally, so an explicit `--max-tokens` was parsed, accepted and discarded. AIME24 read 0.2667 at an applied 8192 against 0.8000 at the requested 32768, and AIME25 0.2667 against 0.7000: a factor of three from a flag that appeared on every command line and reached no sampler. Table 11 is that measurement.

(ii) **A cap larger than the model’s context, which fails every request.** A re-score asked 1.5B checkpoints with `max_position_embeddings: 32768` for 32768 *new* tokens. The server rejected every request, and the scorer reported `acc=nan` with `fail=500/500`. A `nan` in an accuracy column

Table 18: OlympiadBench, $n = 675$, greedy, one run per cell, **trunc** the fraction of generations that reached the cap. At 65536 the two models truncate at the same rate to the precision reported and every problem is graded on both sides, so the comparison is matched on cap, on truncation and on coverage.

model	@32768	trunc	@65536	trunc	graded @65536
Qwen3.8-27B	0.7733	17.8%	0.8089	10.7%	675/675, fail 0
Frontis-MA1-30B	0.7363	20.0%	0.7615	10.7%	675/675, fail 0
gap	+0.037		+0.047		

reads as a score to any reader who does not read the failure column beside it, which is the entire argument for printing that column.

(iii) **A client timeout discarding work the server had already done.** Concurrency was raised $16 \rightarrow 40$ on a 65536 OlympiadBench run while the launcher held a fixed `--timeout 600`. The KV pool was never the constraint; more sequences sharing the cards means each takes longer, and per-request latency crossed the fixed deadline. The server’s access log records 613 generations returned with 200 OK; the client kept 13 of 675 and counted 662 failures. The scorer then reported `acc=0.6154`, `n=13/675`, `fail=662` and exited zero. Roughly a GPU-day of generation produced thirteen usable answers, and the 0.6154 is void: it is an average over self-selected survivors, which are the fastest generations, hence the shortest, hence not a random sample of the benchmark.

Why a warning is not a guard. The scorer did print that its accuracy was over survivors and biased upward. That is correct and it is not enough, because a warning beside a plausible number is read as a number. The abort-on-failure-rate guard that would have stopped this at 98% failures had been written, tested and committed — to a repository the scoring box cannot reach, so the box was still running a copy transferred by hand hours earlier. A fix that exists only in version control does not protect a machine that cannot reach version control. The progress monitor compounded it: it counted server-side completions, which is all a server log can show, and read 90.8% complete for a run that yielded thirteen answers.

The convention we adopt, stated as a requirement on reporting. Every benchmark number in this paper is reported with three things beside it: **the token budget it was taken at, the fraction of generations that hit that budget, and the number of requests that failed.** Any one of the three defects above is invisible without one of them, and each produced a number in the plausible range. The generalisation is that a harness which cannot fail cannot measure: if a run has no path that aborts, then a successful exit carries no information, and the score it wrote is a property of the harness’s defaults rather than of the model.

3.21 The matched frontier comparison at 65536

Sec. 3.16 compares the two frontier candidates at a single cap and records why that is not enough: where a cap binds on both sides, part of the gap is verbosity rather than skill. Sec. 3.17 then shows that OlympiadBench is the one row where the cap was in fact still binding. Table 18 re-runs that row for both models at 65536. It supersedes the statement in Sec. 3.16 that 65536 had not been run for Frontis-MA1-30B.

Table 19: OlympiadBench at a 65536 cap, Qwen3.8-27B. The first three rows are averages over the responses that survived a client timeout and are *not* benchmark scores; the last is the clean rerun at concurrency 24 with a timeout that scales with the token budget. **trunc** counts *graded* generations that reached the cap, so it is a count over the survivors in the first three rows.

read	accuracy	graded	failed	trunc
survivors, first read	0.6154	13/675	662	—
survivors, further on	0.8717	265/675	410	0
survivors, furthest on	0.8891	613/675	62 (9.2%)	1
clean rerun	0.8089	675/675	0	72 (10.7%)

The ranking survives a matched, less-truncated budget. At 32768 the gap is +0.037 with the two models truncating at 17.8% and 20.0%; at 65536 it is +0.047 with both at 10.7%. The gap does not close when the budget doubles — it widens slightly — and it widens at the point where the truncation asymmetry that could have explained it has gone. This is what a single-cap comparison cannot establish. At one cap, “the stronger model is simply the terser one” is an alternative account of the entire gap, and no amount of care about the grader touches it, because the two models are not being asked the same question when one of them is cut off more often. Two caps, with the truncation rates converging to each other and the ordering unchanged, is the observation that rules it out.

The larger model is still on the slope. Across three caps Frontis-MA1-30B reads 0.6237 at 16384 (33.3% truncated), 0.7363 at 32768 (20.0%) and 0.7615 at 65536 (10.7%). It is still climbing, and so is Qwen3.8-27B. Neither 65536 figure is a measurement of capability; both are capability under a 65536 budget, and both remain lower bounds while 10.7% of generations reach the cap. What is now established is the *ordering*, at a budget where the verbosity confound has been reduced rather than assumed away.

Scope. One run per cell, greedy. Both rows inherit the grader caveat of Sec. 3.12: 94 of the 675 problems state more than one gold answer in a single answer string and are marked wrong however the model answers. That caveat applies equally to both models, so it moves both accuracies and not the gap.

3.22 Survivor bias is not conservative: an average over survivors is biased upward

The OlympiadBench figure at 65536 was read three times before it was measured once, and the three wrong reads are worth more than the incident that produced them, because they quantify a bias that is ordinarily assumed to run the other way. The mechanism is the one Sec. 3.20(iii) reports: concurrency was raised while the client held a fixed timeout, so generations the server had already completed were discarded client-side, and the scorer averaged over whatever survived. Table 19 places those survivor averages beside the clean rerun.

The bias is upward, and it is larger than the effect being measured. The best survivor read, 0.8891 over 613 of 675, overstates the clean 0.8089 by +0.08. The effect that run existed to measure is the cap effect of Sec. 3.17, +0.036. The bias is more than twice it. A survivor average

is therefore not a noisier version of the score with the same centre; it is a different quantity, and quoting it would have more than doubled the headline in the direction that flatters the model.

The direction is predictable, which is what makes this methodology rather than an anecdote. What the client discarded were the requests that took longest. Longest means most tokens, and on a mathematics benchmark the longest responses are the ones on the hardest problems — the same correlation that makes truncation itself a difficulty-biased filter. Removing them removes hard problems from the average, so the average rises. The truncation column shows the same selection from the other side: the 613-survivor read contains exactly *one* truncated generation against 72 in the complete run, because a generation long enough to reach the cap was also slow enough to be discarded. A read that loses the slow responses is structurally unable to see the cap effect it was launched to measure.

Against the intuition that dropping data is safe. The reflex is that a partial result is a conservative result: fewer samples, a wider interval, the same centre. That holds only when the drop mechanism is independent of the outcome, and here it is anti-correlated with the outcome by construction. The failure is also silent. All three survivor rows are plausible OlympiadBench numbers, all three came from processes that exited zero, and the 0.8891 row has a 9.2% failure rate that a reader scanning for problems would not consider disqualifying. We therefore treat the completion count as part of a score’s identity rather than as diagnostics beside it, which is the reporting convention of Sec. 3.20, and we make a high failure rate abort the scorer rather than warn: a warning printed next to a plausible number is read as a number.

A correction this forces on Sec. 3.20. That subsection reports the incident as a GPU-day of generation yielding thirteen usable answers. The mechanism is right and the 0.6154 is void either way, but the magnitude is wrong. The output file accumulated several launches under one reused tag; the 13/675 line was the earliest of them and the same file’s furthest-completed launch reached 613/675. The loss was overstated by reading one line of a file that other processes were still writing. A reused output tag makes a log a mixture of runs, and a single `grep` of such a file is not a measurement.

3.23 An unrouted 30B LoRA control, and what it is not

What this run is, read from its resolved configuration. `lora30b` trains a LoRA adapter on Frontis-MA1-30B with GRPO on eight A100s. Its resolved configuration reads `group_routing: null` and `token_routing: null`. **The run has no routing of any kind. It is the matched control for a routed arm at 30B, not an instance of the method.** We state this before the numbers because the run was described internally as the method for several days on the strength of its name. A name is not a configuration, and the only thing that settles what a run tested is the switch in the config it resolved.

The conclusion is approximately neutral. The peak rung is step 149 at both caps, and the base-to-peak gap is +0.032 at 4096 and +0.018 at 16384. At 16384, with $p \approx 0.88$ at $n = 500$, the standard error of one accuracy is 0.0145 and a difference of two runs carries about 0.021, so +0.018 sits inside one standard error of the difference. At 4096, with $p \approx 0.67$, the same figures are about 0.021 and 0.030, so +0.032 is roughly one standard error there too. Neither cap supports calling the peak an improvement. At 16384 the base is not even the lowest rung: four of the eight adapter

Table 20: `lora30b` adapter checkpoints against the adapter-free base, held-out MATH-500, $n = 500$ per cell, greedy, every cell graded 500/500 with zero request failures. The same nine rungs are scored at two caps. `trunc` is the fraction of generations that reached the cap. This is a control ladder: no rung has routing enabled.

step	cap 4096		cap 16384	
	accuracy	trunc	accuracy	trunc
base, no adapter	0.6520	41.2%	0.8820	13.8%
24	0.6600	41.0%	0.8680	15.0%
49	0.6660	39.2%	0.8720	15.0%
74	0.6720	40.2%	0.8880	12.8%
99	0.6700	39.8%	0.8800	13.2%
115	0.6680	38.6%	0.8740	14.2%
124	0.6720	40.4%	0.8880	12.6%
149	0.6840	39.4%	0.9000	11.4%
174	0.6760	40.8%	0.8820	13.2%
base to peak	+0.032		+0.018	

checkpoints score below it, the worst by 0.014. **LoRA on this base is approximately neutral on held-out MATH-500**, and that is the claim we make for it.

The gap shrinks as the cap grows, and the truncation column says why it might. +0.032 at 4096 becomes +0.018 at 16384. Truncation falls along the ladder in both columns, and the base is the more truncated end of it in both: 41.2% against the peak’s 39.4% at 4096, 13.8% against 11.4% at 16384. Part of the apparent gain is therefore not accuracy but the base being cut off more often than the peak adapter, which is to say the adapter becoming more concise as training proceeds — the opposite of the length growth the routed 1.5B arms produced (Sec. 3.13). A gain that shrinks as the confound shrinks has the shape of an artefact, and extrapolated to zero truncation this one plausibly closes. Three rungs are being scored at 65536 to test that: if the gap returns at +0.018 or larger where truncation is near zero, the truncation account is wrong and the effect is real. That is a prediction and it is not a result.

What survives on the other side, recorded because it is not settled. The *shape* replicates across two budgets differing fourfold with truncation differing threefold: the same peak at step 149, the same downturn at 174, and a near-monotone ordering of the rungs in between. The rise from step 24 to step 149 is present in both columns (0.6600 $\rightarrow$ 0.6840 at 4096, 0.8680 $\rightarrow$ 0.9000 at 16384). Noise around a constant does not usually arrive sorted twice. We record that as suggestive and decline to claim it, because the significance calculation above is the binding constraint and it says no.

What the ladder does settle: the feared degradation is absent. `lora30b` inherits `eps_clip: 0.4` and `lr: 1.0e-05` from an upstream demo recipe, and the aggressive recipe of Sec. 3.1 cost `step0d 0.194` absolute on this same benchmark, monotonically, under full fine-tuning of a 1.5B. Nothing of that shape appears here at either cap. That is why the checkpoints were scored rather than the run restarted, and it is the negative claim the ladder was launched to test. It does not transfer the 1.5B finding either, because two things changed at once — the scale, and full fine-tuning to LoRA — and adapters are conventionally trained one to two orders of magnitude above full

fine-tuning learning rates, which makes this learning rate conservative by LoRA standards rather than aggressive.

3.24 Scope of the routing claims, stated as a limitation

Every routing measurement in this paper is at 1.5B. `ctx`, `ctxpc`, `ctx2`, `rnd` and `step0j` all train Qwen2.5-1.5B-Instruct. The null against the matched control (Sec. 3.9), the credit-signal analysis (Secs. 3.10 and 3.11), the length threshold (Sec. 3.13), the termination probe (Sec. 3.15) and the collapse and its timing (Sec. 3.19) are all claims at that scale. None of them is evidence about routing at any other scale.

No routed arm at 30B has been run. The only 30B training run reported here, `lora30b` of Sec. 3.23, has routing disabled and is the control half of the comparison the paper’s claim needs; the treatment half does not exist. Sec. 3.23 therefore bears on plain GRPO with a LoRA adapter at 30B and on nothing else, and no reading of it as a routing result at a fixed strong base is supported. The frontier work of Secs. 3.16, 3.17 and 3.21 selects and characterises that base; it does not measure a delta on it. A delta at a fixed strong base is the paper’s stated goal and it is currently half-measured.

The two stabilisers have never been used by an experiment. The zero-mean advantage correction (`zero_mean`), which re-centres the advantage field after all routing writes so the batch mean is restored, and the exclusion of non-terminating rows from the supervised write (`exclude_truncated_from_sft`) are both implemented in the actor, both default to off, and are covered by a mutation suite that kills 39 of 39 mutants. No run reported in this paper enables either. They are code, not results: this paper makes no measurement of their effect, and the mechanism arguments that motivated them stand as arguments.

3.25 The reporting requirement, and a fourth harness failure that has now recurred

Sec. 3.20 records three ways this harness produced a number that was not a measurement, and Sec. 3.18 measures how far the same pipeline moves when it scores the same checkpoint twice. Read together they settle two things every accuracy in this paper depends on: what has to be printed beside a score, and which scores may be compared at all. This subsection states both as rules and adds a fourth failure mode — the only one of the four that has now been observed more than once, and on more than one machine.

Comparability is set by truncation, not by a single noise band. The accidental duplicate run of Sec. 3.18 is the measurement that fixes this: at $\approx 10\%$ truncation the same checkpoint reproduces to 0.002–0.010, and at $\approx 97\%$ truncation it moves by 0.024–0.026. Same checkpoints, same matched 16384 cap, same grader; the only thing that differs is how much of the batch the cap is cutting off, and the reproducibility differs with it by an order of magnitude. There is therefore no single noise floor for this suite to quote, and quoting one would be wrong in both directions at once — too permissive at 10% truncation and far too strict at 97%. The operating rule is the one Sec. 3.18 states and this paper applies throughout: **accuracies taken above $\approx 80\%$ truncation are not comparable to one another**, and where two such points differ we read the truncation channel instead. The post-transition points of Sec. 3.19 are the case that forced the rule, and Sec. 3.23 is the case where it is comfortably satisfied on both ladders.

Table 21: Three occurrences of one failure mode: a generation cap larger than the model’s context window. In every row the server rejected every request, the scorer wrote a **nan** accuracy with a full failure count, and the run was recorded as complete. The collaborator’s context windows are as recorded in that sweep’s models rather than re-verified here.

machine	model	context	new tokens asked	reported
ours	ctx2/rnd 1.5B ckpts	32768	32768	acc=nan, fail=500/500
collaborator, 4×H200	Qwen2.5-32B-Instruct	≤ 32768	32768	acc=nan, fail=N/N, DONE
collaborator, 4×H200	Qwen2.5-Math-7B-Instruct	4096	32768	acc=nan, fail=N/N, DONE

(iv) **A cap larger than the model’s context, recurring across machines.** Sec. 3.20(ii) reports this mode once, on our own re-score of 1.5B checkpoints. It has since occurred twice more, inside a single sweep on an independent four-GPU H200 box belonging to a collaborator, running a revision of the harness that predates the failure-rate abort. That sweep passes a 32768 cap unconditionally; two of the three models it scored have context windows at or below 32768, one of them 4096. The server rejected every request. Both runs finished in minutes (5.5 and 3), both reported **acc=nan** with **fail=N/N** on every benchmark in the suite, and the sweep recorded both as DONE. Table 21 lists the three occurrences.

The one model in that sweep that did produce numbers is excluded rather than reported. Two independent reasons, either of which is sufficient. Its identity is inferred from the ordering and duration of the runs rather than read from a configuration, and this paper has already been wrong about what a run was on exactly that kind of evidence (Sec. 3.23). And its OlympiadBench row discarded 83 of 675 requests client-side, 12.3%, which makes it the survivor average of Sec. 3.22 rather than a score — the same construction that overstated our own 65536 figure by +0.08. We record the sweep as evidence about the failure mode and take no accuracy from it.

The durable fix is a clamp at startup, not a guard at runtime. A failure-rate abort already exists in this codebase and would have stopped all three runs of Table 21 before either wrote a number: each fails every request, which is as far past any threshold as a run can get. It stopped none of them, because in each case the machine was running a revision older than the guard. That is now the second time a guard which was written, mutation-tested, committed and pushed failed to protect a machine, and the two causes have the same shape — once the scoring box had no credentials with which to fetch the repository (Sec. 3.20), once it had simply not pulled. **A runtime guard protects only the revisions that have pulled it.** The quantity this mode turns on, however, is not a runtime observation at all: `max_position_embeddings` sits in the model’s own `config.json` and is readable before a single GPU is allocated. A scorer that reads it and clamps the requested cap, or refuses to start, is correct on every revision that runs afterwards rather than on every revision that has caught up. The clamp is now implemented in the scorer and covered by a mutation suite: it reads the context from the model configuration, reserves room for the prompt, and prints what it changed and why, while an unreadable or absent configuration yields *unknown* and never clamps, since not knowing a context is not evidence that it is small. It postdates every run reported in this paper, so no number here was protected by it, and the three rows of Table 21 are what it was written against.

Table 22: The liveness probe’s two controls, measured against the training run’s own rollout server while it was training. A working difference metric must report the first row at or near zero and must separate the two rows. The original probe does neither. The repaired probe is the same six prompts on the same server minutes later; its signal column is exactly zero on identical weights because the comparison is now between the same tokens in the same contexts.

probe	control	compared quantity	value	verdict
original	base vs. base (identical weights)	$\max \Delta \ell $	0.978	live
original	adapter vs. base	$\max \Delta \ell $	1.007	live
repaired	base vs. base (identical weights)	$\text{mean } \Delta \ell $	0.00000	not live
repaired	adapter vs. base	$\text{mean } \Delta \ell $	0.0446–0.0534	live

What a score has to carry, restated with what these subsections add. Sec. 3.20 requires three things printed beside every benchmark number: the token budget, the fraction of generations that reached it, and the number of requests that failed. Two additions follow from the subsections above. First, the *denominator* is printed as graded over total rather than left to be recovered by subtracting the failure count, because Sec. 3.22 shows that an average over the survivors of a partial run is not a noisier estimate of the same quantity but a different and upward-biased one, and a number whose denominator a reader must reconstruct is a number whose denominator a reader will not check. Second, comparability is gated on truncation by the rule above, so a pair of accuracies is a comparison only when both sit below the threshold. Each of the four failure modes recorded in this paper produced a plausible number from a process that exited zero, so none of them is detectable anywhere downstream of the score; these columns are the only place they are visible.

3.26 A fifth failure mode: a guard that could not fail

The four modes above all produce a plausible *number* from a broken process. This one is worse in kind: it is a *check* that was installed to catch a specific failure, that ran on schedule, that reported a healthy verdict every time — and that would have reported the same healthy verdict if the thing it was watching for had been happening continuously. It was found by pointing it at a case whose answer is known in advance, which is the only way this class of defect is ever found.

The check, and why it was there. An arm trained with a LoRA adapter is served by the same endpoint as its base model, addressed by model id. If the id fails to resolve, the server answers HTTP 200 with the *base* model, and the arm’s accuracy is then the base model’s accuracy wearing the arm’s label. The periodic evaluation therefore carries an adapter liveness probe: six fixed prompts, one greedy completion from the adapter id and one from the base id, per-token logprobs from each, and a verdict from $\max_i |\ell_i^{\text{adapter}} - \ell_i^{\text{base}}|$ over the $6 \times 32 = 192$ generated positions, against a threshold of 10^{-4} .

The negative control, which is the whole result. We pointed that probe at the base model with no adapter, so the two things being compared were the same weights and the correct answer is *not live, difference zero*. Table 22 gives what it returned, beside a real adapter measured minutes earlier on the same server. The probe reported the base model as **live**, at a value that does not separate from a genuinely different set of weights. Every liveness verdict this project had recorded was therefore uninformative, and in particular the two that had been read as confirming an arm was not silently scoring its base.

The cause is that the two sides were never compared at the same tokens. The probe generated *independently* from each side and then subtracted the two logprob streams by position. Greedy decoding on a continuously batching server is not reproducible: the same weights produced different text on one of the six prompts, and after the argmax paths diverge, position i holds a different token on each side, so the subtraction compares the logprob of one token under one model with that of another token under the other. Divergence happens only at near-ties, where the losing branch sits near $\log \frac{1}{2} = -0.693$ and the winning branch near 0; the maximum therefore lands near 1 regardless of the adapter. That is a saturation, not a measurement, and it explains the observation that first drew suspicion — two evaluation points fifty steps and fifty adapter versions apart agreeing to five decimal places, 1.04127 and 1.04126.

The run’s own record already contained the refutation. At steps 50 and 100 the probe reported a text-divergence fraction of 0.167 — exactly one prompt in six, the same same-weights flip we reproduced — with $\max |\Delta \ell| = 1.041$. At step 150 no prompt diverged, and the reported difference fell to 0.117: a ninefold drop on an adapter that had only trained further. The series was tracking a coin flip in the decoder, not the weights.

A second defect the control exposed, which no threshold could have absorbed. Scoring one fixed 61-token sequence on the base route 24 times returned exactly two distinct logprob vectors, 22 and 2 times, differing by 0.605 nats at a single position. It is not adapter leakage: the deviant vector is equidistant from the adapter and from the modal base. So the same-weights noise on this server is of the same order as the adapter signal, and the shipped threshold of 10^{-4} was never usable no matter how the difference was computed.

What the repair has to do, and how it is held in place. The probe now generates nothing. It *scores* a fixed token sequence under each set of weights, so every compared position is the same token in the same context; it scores each side twice and takes the smallest cross-pair difference as signal and the largest within-side difference as an in-band negative control, so the server’s own nondeterminism is measured beside the signal on every call rather than assumed; and it decides on the mean over positions rather than the maximum, which no single near-tie can carry. Pooled over six prompts and 465 positions, base against base is exactly 0 on seven independent sweeps while the adapter differs at 378 of the 465 positions with a pooled mean of 0.0525. The negative control is now a test rather than something someone has to think to run, and eight distinct mutations of the repaired probe — including comparing a distribution with itself, defaulting the verdict when nothing was compared, and reporting the configured adapter name rather than the served one — are each caught by it.

One calibration item remains open, stated as a number. The threshold is still 10^{-4} while the largest same-weights pooled mean we have measured is 0.0095 and the weakest real adapter signal is 0.0446. In 39 negative controls against the intermediate form of the repair, one came back live at 0.0095. The final form suppresses this — 0 of 27 — because the minimum over cross-pairs finds a clean pairing whenever only one scoring lands on the deviant path, but two scorings of the same side landing there together is not excluded, and at that point the threshold is the only thing standing between the base model and a verdict of live. The usable window is bounded below by 0.0095 and above by 0.0446.

min. cluster size	clusters	behavioural	size-matched random	prompt-gradient
2	23	-0.00122	-0.00138	+0.00134
3	18	-0.00000	-0.00097	+0.00357
5	8	+0.00579	-0.00329	+0.01155

Table 23: Mean pairwise cosine between per-cluster gradients at `globalstep199`, 140 informative groups. The hypothesis requires the behavioural column to sit *below* the random column. It sits above it at every setting.

The cost of the check, and what does not explain it. The evaluation’s cost is rising: 104.2, 139.5 and 177.4 wall seconds at steps 50, 100 and 150, or 7.8%, 10.3% and 12.9% of training throughput. The obvious explanation, that generations lengthen as training proceeds, is false here: mean generated length is flat at 2289, 2336 and 2287 characters, the finish-reason mix is identical at 61 stopped and 3 truncated, all 64 problems graded with no failures at every point, and the log records no aborted requests. Nor is the run slowing: implied step time moves only from 26.7 to 27.5 seconds. The growth is confined to the benchmark generation phase (88.7, 123.9, 155.4 seconds) and is not the liveness probe, which after the repair is prefill-only and cannot grow with completion length at all. Three points cannot separate a linear trend from a saturating one, and step count and wall-clock are collinear here, so we do not claim a mechanism; we record that nothing measured so far bounds it, and that on the observed slope the cadence reaches the whole of training throughput near step 1900. The remedy that follows from this is a ceiling on projected cost rather than a change to what an evaluation measures: project each point’s duration from the previous point and the measured step time, and skip the point with a distinct status code when the projection exceeds a configured fraction of throughput, so that the gap is visible on the curve instead of silent.

3.27 Not yet measured

Stated explicitly so the gap is not mistaken for an omission: whether training on the solved branch helps or costs entropy (Sec. 3.5 locates the mass but does not settle this); any arm in which routed units *gain* a signal rather than only losing one; and the harness-destination arm of (4). Until a target is supplied, every routed arm is a gradient-deletion ablation and is labelled as one.

3.28 The interference probe returns a null, and the pre-registered gate closes

The method’s central mechanism is that behavioural subpopulations inside one task want conflicting adapter updates, so that partitioning them and giving each its own adapter removes a conflict that a single shared adapter must average away. That claim is testable before any adapter is trained, and we fixed the decision rule before computing anything: proceed only if the behavioural partition’s mean pairwise gradient cosine falls *below* a size-matched random partition’s by more than the bootstrap interval; treat parity with the random partition as evidence that the clusters are noise and stop.

We dump exact per-cluster LoRA gradients at `globalstep199` over 140 informative groups, and compare four partitions on the same batch, the same checkpoint and the same gradients. Table 23 reports the result at three settings of the clustering’s minimum cluster size.

The sign is opposite to the hypothesis at all three settings, and the magnitudes are within noise of zero throughout. Conflict rates tell the same story more plainly: the fraction of cluster pairs with negative cosine is 0.502, 0.477 and 0.500 for the behavioural partition against 0.534, 0.484 and

0.500 for the random one, and one half is what a coin gives. Prompt-gradient clusters, which require no rollouts at all, are consistently the *least* conflicting of the three, which removes the secondary argument that behavioural features are what make the partition work.

Two readings are available and we take the conservative one. The clustering is not degenerate: it finds between eight and twenty-three clusters depending on the setting, so this is not a failure to separate anything. What it does not do is separate *gradients*. Cancellation sits between 0.24 and 0.41 for both the behavioural and the random partition, so conflict exists in the batch; it is simply isotropic with respect to any labelling we can compute. The honest summary is that there is very little structured gradient there to partition, and the null is not an artefact of the sketching used to make the comparison affordable. Thirty-two gradients were also stored exactly, and across their 496 pairs the largest disagreement between sketched and exact cosine is 0.0314 against a resolution floor of 0.0331, so the sketch resolves what it is being asked to resolve with margin to spare. The exact mean pairwise cosine is -0.000506 , agreeing with the sketched figure. Two further measurements point the same way: the per-cluster gradient norms span only about a factor of two, so no cluster dominates the average, and the prompt-side negative log-likelihood gradient is roughly 65 times larger than any of them, which is where the structure in this batch actually lives.

The honest summary stands: there is very little structured gradient there to partition, and the pre-registered rule therefore closes: we build no per-cluster adapters on this signal.

The weaker use of the same clustering as a *targeting* key requires only that clusters differ in how often the model succeeds, rather than that their gradients separate, and we test it on the same batch. It fails as well. Taking the fraction of variance in per-group solve rate explained by the labelling, the behavioural partition reaches 0.179, 0.160 and 0.071 at the three settings against 0.189, 0.190 and 0.073 for size-matched random partitions drawn over 4000 permutations, with p between 0.45 and 0.67. It explains *less* than chance at every setting.

That result is worth stating in the form that shows why the control is not optional. Cluster mean solve rates range from 0.21 to 0.88, and the labelling accounts for eighteen percent of the variance; reported alone, both numbers read as a finding. They are what feature-blind labels of the same sizes produce, because with twenty-two clusters over a hundred and twelve points small groups inflate the statistic mechanically. The control is what separates a discovery from an artefact of partition size, and here it removes the whole effect.

3.29 Validating the graders against published numbers

No published score exists for our base model on either of our evaluation sets, so neither grader had ever been checked against anyone else’s run. We close that with two proxies chosen so that the target is fixed before the measurement.

Our OlympiadBench file is the 675-row open-ended English competition subset, and a released math model’s evaluation parquet contains that split at exactly 675 rows, so the problem sets are identical rather than similarly named. Scoring QWEN2.5-7B-INSTRUCT through our own grader gives 0.394 with a Wilson interval of $[0.358, 0.431]$ against a published 39.7: the grader agrees with the literature to a third of a point. On MATH-500 our base model scores 0.824, $[0.788, 0.855]$, against a published band of 84.0 to 84.6; the interval covers the target, and the point estimate sits at its low end, within the roughly 2.3-point spread that separates two published evaluations of the same released checkpoint.

The third check did not succeed, and its failure is more informative than its success would have been. A frontier 27B model reports 90.3 on our code benchmark’s name, so we scored it through our grader and obtained 0.251 to 0.269 across seeds. That gap does not measure our grader: 131 to 136 of 175 generations, about 76%, exhausted the token budget and were graded as failures, and

benchmark	problems	accuracy	Wilson 95%
OlympiadBench, all	675	0.470	[0.432, 0.507]
<i>search half</i>	338	0.482	[0.429, 0.535]
report half	337	0.457	[0.405, 0.510]
LiveCodeBench (v6 slice)	175	0.309	[0.245, 0.381]

Table 24: Baseline arm at `globalstep149`. Every problem graded, none failed. The math figure is a lower bound: 94 of 675 problems carry several gold answers in one string and exact-match scores them wrong regardless of the response.

43% still truncated at a cap raised to 24,576. Raising the cap resolves it, and the resolution is the measurement. At 16,384 tokens the model scores 0.251 with 76% of generations truncated; at 61,440 it scores 0.749, [0.679, 0.807], with truncation down to 23.4%. Nothing about the grader changed between those two runs. The entire difference is the token budget.

Conditioning on completion settles it more sharply than that bound does. At the canonical cap, *all* 131 failures were truncations and the grader rejected nothing that finished. At 61,440, of the 134 generations that terminated normally, 124 passed: 92.5%, against a published 90.3. Conditioning on completion selects the easier problems, so 92.5% is not an unbiased estimate of the model’s score and we do not report it as one. It does answer the binary question the check was for. Nothing our grader saw finish was wrongly rejected, and what it rejected, it rejected for running out of tokens. We therefore treat the code path as externally corroborated, alongside its internal checks: gold self-verifying on all 175 problems, a known-wrong set failing on all of them, and every mutant killed.

The general point outlives this particular check. A code benchmark score is not interpretable without the generation budget that produced it, and a difference of four times in that budget moved the same model on the same problems through the same grader from 0.251 to 0.749. Almost no published card reports its cap. Our own baseline truncated on no problems at all, so its score is a property of the policy, but that is a fact we had to measure rather than assume.

3.30 Baseline scores

Table 24 gives the vanilla arm at `globalstep149`, which is the reference every later comparison is relative to. The split is committed and content-pinned, and *every arm decision reads the search half*, so 0.482 rather than 0.470 is the operative figure; the full set and the reserved half are shown once here and the reserved half is not consulted again until the final write-up. Both sit in wide separable bands: the math score against roughly 0.83 for a frontier 27B, and the code score against the 90.3 discussed above. Neither is floored and neither is saturated, which is the property that makes an arm difference detectable at all.

The code run truncated on no problems at all and was not budget-bound, so its 0.309 is a property of the policy rather than of the cap — the contrast with the frontier model above is the clearest illustration in this paper of why a generation budget must be reported alongside any code score.

3.31 Correction to Sec. 3.29: the canonical code check ran at 2000 tokens, not 16384

Sec. 3.29 reports the third grader check — a frontier 27B model scored through our own code grader — and states that the low score was taken “at 16,384 tokens”, and that “a difference of four times

Table 25: The third grader check, re-derived from the results JSON of each run. The canonical row is ten seeds at the benchmark’s own default cap; the two raised caps are one seed each. `trunc` is the fraction of the 175 generations that hit the cap, read from `truncation_rate` in the same file. Runs `canonical`, `noncap` and `noncap2` under `runs/validate_v3/`. The canonical row’s accuracy and interval columns are ranges across its ten seeds — the lowest and highest point estimate, and the lowest lower bound and highest upper bound — not one interval.

cap	seeds	accuracy	Wilson 95%	trunc
2000	10	0.2514–0.2800	[0.193, 0.351]	74.9–77.7%
24576	1	0.6457	[0.572, 0.713]	42.9%
61440	1	0.7486	[0.679, 0.807]	23.4%

in that budget” carried the same model from 0.251 to 0.749. Both figures are wrong. Re-derived from the run’s own artifacts, the canonical cap was 2000 and the budget ratio is about $31\times$. The correction makes the subsection’s argument stronger, not weaker, which is the reason to state it precisely.

Every one of the ten canonical seeds (`canonical.livecodebench_v6.seed0..9.json` under `runs/validate_v3/`) records `params.max_tokens: 2000`, and the launcher that produced them (`harness4/run_v3.sh`) carries MAXTOK defaulted to 2000, which is the benchmark’s own upstream default rather than a value this project chose. No run of that model at 16,384 exists. Table 25 restates the ladder from the JSON.

Three numbers move and one does not. The across-seed range is 0.2514 to 0.2800, not the 0.251 to 0.269 recorded: the top of the range is seed 8 at 0.2800, and quoting the range short of its maximum understates the seed spread by a fifth. The budget ratio between the two ends of the ladder is $61440/2000 = 30.7\times$, not four. The truncation figure at the canonical cap is 131 to 136 of 175, which is what Sec. 3.29 already says and which is unaffected. So is the whole conditional-on-completion argument, which we re-derived per generation from `finish_reason` in the records files rather than from the summary counters: at the canonical cap 39 generations terminated normally and *all* 39 passed, so the grader rejected nothing that finished; at 61440, 134 terminated normally and 124 passed, 0.9254, which is the 92.5% already reported.

Why the corrected version is the stronger one. At 16,384 the reading would have been that a frontier model needs four times a generous budget to be scored fairly. At 2000 it is that the benchmark’s *own published default cap* truncates three quarters of a frontier model’s generations and costs it half a point of accuracy, and that a card reporting a score under that default is reporting a property of the default. That is the general claim Sec. 3.29 makes, and the corrected cap is the evidence for it. The lesson for our own reporting is the one this paper keeps relearning in a new place: the cap was read out of a summary rather than out of `params.max_tokens` in the row, and the row had carried the right number the whole time.

3.32 Corpus saturation: the baseline’s first corpus left 72% of its groups silent

The A0 baseline arm (Qwen2.5-32B-Instruct, LoRA rank 32, GRPO, 8 prompts $\times$ 8 samples per step, `group_routing: null`) was first launched on a MATH corpus and then relaunched on decontaminated DeepMath with every other setting held fixed. That accident is a controlled measurement of how much of the silent channel is a property of the *corpus* rather than of the

Table 26: The silent channel under a corpus change at fixed model and configuration. Each row is one launch; `n` is its logged steps. The MATH row is `runs/a0_math_MATHCORPUS_superseded/train.log.1788330321` and `.1788333052`; the DeepMath rows are `runs/a0_math/train.log.1788336152 + .1788344500`, `.1788371484` and `.1788377789`. The parser matches each key with the cell delimiter in front of it, because `solved_group_fraction` is a substring of `unsolved_group_fraction` and a bare match for the first also matches every row of the second — the aliasing that forced the retraction recorded on 2026-08-31.

corpus	launch	<i>n</i>	silent	solved	unsolved	informative
MATH	<code>.1788330321+.1788333052</code>	88	0.7188	0.6321	0.0866	28.1%
DeepMath	<code>.1788336152+.1788344500</code>	234	0.5043	0.3365	0.1677	49.6%
DeepMath	<code>.1788371484</code>	509	0.5206	0.3480	0.1726	47.9%
DeepMath	<code>.1788377789</code>	212	0.5047	0.3314	0.1733	49.5%

method, and it is the reason the arm was moved. Table 26 reads the four `*_group_fraction` series the actor logs unconditionally on every step, out of the training logs themselves.

The rollout dumps say the same thing without going through a logged metric. Counting a group as unanimous when every reward in it is equal, directly over the exported rollouts: on the MATH corpus 193 of 261 groups across two probes are unanimous, 0.7395, with the step-24 probe alone at 102 of 133, 0.767; on DeepMath 533 of 1025 groups across eight probes are unanimous, 0.5200, with no single probe exceeding 0.570. Two independent instruments — a per-step metric the trainer emits and a per-group recount of the dumped rollouts — agree to within the sampling difference between them.

What saturation does to the update, stated in the quantity that matters. On MATH the median gradient norm over the 88 steps is 1.82×10^{-4} and 11 of the 88 steps have a gradient norm of exactly zero: on one step in eight the entire batch was unanimous and the update was a no-op. On DeepMath the median gradient norm is 5.18×10^{-4} , $2.8\times$ larger, and *no* step in 234 has a zero gradient. Mean train reward moves the opposite way, 0.7923 on MATH against 0.6034 on DeepMath, which is the point: the corpus that produced the higher training reward is the one that was teaching the model almost nothing.

Why this belongs in a results section and not in an operations note. A run on a saturated corpus does not fail. It exits zero, its loss is finite, its reward is high and rising, and its logs contain no error — and three quarters of every batch contributes exactly nothing to the update. This is the same failure shape as Sec. 3.1, one level further back: there, train reward mis-ordered checkpoints; here, train reward is high *because* the corpus is already solved, and the higher number is the worse arm. The practical rule we adopt from it is that a corpus is qualified for an arm by its unanimity rate measured on that arm’s own model, before the arm is launched, and not by its reputation for difficulty.

Scope. One model, one pair of corpora, one configuration; the MATH row is 88 steps against 234–509 for DeepMath, and the two corpora were run at different points of the same day on the same box. The claim is a corpus difference at this model scale, not a property of either dataset in general.

Table 27: The unsolved share of the silent channel, at the operating point the earlier table measured and at the one now being trained. The first two rows are Table 6 restated; the last three are the three A0 launches of Table 26, whose **unclassified** series is exactly 0.0000 on every one of their steps, so unsolved/silent and unsolved/(solved + unsolved) are the same number here and the share is well posed.

task	model	n	unsolved, all groups	unsolved share of channel
GSM8K	1.5B	98	0.045	12.5%
MATH	1.5B	11	0.255	60.9%
DeepMath	32B	234	0.1677	33.3%
DeepMath	32B	509	0.1726	33.2%
DeepMath	32B	212	0.1733	34.4%

3.33 The unsolved branch is a third of the silent channel on the arm actually trained

Table 6 reports that the composition of the silent channel inverts with task difficulty, reaching 60.9% unsolved on MATH, and the plan for a data supplier is sized on that figure. Both columns of that table are Qwen2.5-1.5B-Instruct, and the MATH column is $n = 11$ batches. Re-measured on the arm this paper actually trains — Qwen2.5-32B-Instruct on decontaminated DeepMath — the unsolved share is about half of it.

Three independent launches agree, so this is not a window effect. The three DeepMath rows are three separate launches of the same configuration on three different logs, totalling 955 logged steps, and their unsolved shares are 33.3, 33.2 and 34.4 percent. Within the longest of them the share is stable rather than drifting: over the second half of the 234-step series it is 34.0% against 33.3% over the whole.

The confound is stated rather than buried, because it is not separable here. Table 6 holds the model fixed and varies the task. These rows vary both, 1.5B to 32B and MATH to decontaminated DeepMath. A larger model solves more, so a smaller share of its silent channel is unsolved by construction, and nothing in this measurement separates “DeepMath is easier for this model” from “scale shrinks the unsolved branch”. It is reported as one number and not as an attribution.

What it changes for the supplier axis, which is the reason to report it. The supplier is the only consumer that can act on the unsolved branch, so the size of that branch is the size of the lever. At 64 groups per batch, 0.167 is about 10.7 groups per batch with no target of any kind, against the 16 that a 60.9% reading implies. The lever survives — a sixth of every batch still reaches the update with nothing to learn from — and it is worth about half what the quoted figure claims on the arm being trained. No statement sized on 60.9% may be carried onto a DeepMath result.

3.34 The first periodic evaluation points of the A0 baseline

The baseline arm now scores a held-out slice during training. Table 28 gives the points it has produced, read from `runs/a0_periodic/step{50,100,150}/results.json` as of 21:00 UTC on 2026-09-02, while the run continues.

Table 28: A0 periodic evaluation, OlympiadBench search half, 64 problems, greedy, one sample. **adapter** is the LoRA version the rollout server actually resolved and served for that point, recorded in the row rather than configured. Every point graded 64 of 64 with no request failures and 3 truncations. **These are not accuracies** and must not be placed beside any other number in this paper: see the budget columns.

step	adapter	trend score	Wilson 95%	budget	headline budget
50	a0_math-v51	0.1094	[0.054, 0.209]	2048	16384
100	a0_math-v101	0.1250	[0.065, 0.228]	2048	16384
150	a0_math-v151	0.1406	[0.076, 0.246]	2048	16384

The budget is not the headline budget, and the series is named so that it cannot be mistaken for one. These points are generated by the *training* rollout server, which publishes `context_length`: 4096 in its own server info, so the evaluation runs at 2048 new tokens against the 16,384 every headline number in this paper was taken at. Each row therefore carries `budget_matches_headline`: 0 and `not_comparable_with_headline`: `true`, and the quantity is published as `trend_score` rather than `accuracy` — there is no accuracy series in that namespace at all, so there is nothing a reader can line up against a headline accuracy by name. Sec. 3.31 is what that convention is defending against: on this project’s own measurements a budget change alone moved one model on one problem set through one grader from 0.2514 to 0.7486.

What the three points do and do not support. They rise monotonically, 0.1094 to 0.1406. At $n = 64$ the three Wilson intervals overlap almost entirely, so **no trend is claimed**: the interval at step 150 contains the point estimate at step 50. What the series does establish is the thing it was built for. The adapter version advances exactly one per training step (51 at step 50, 101 at 100, 151 at 150), and it is read from the server’s response rather than from the configuration, so the weights being scored are provably the weights being trained — the failure this project measured at `globalstep149`, where an evaluation could not distinguish an arm’s adapter from its base model, cannot silently recur here. Truncation is constant at 3 of 64 across all three points, so the scores are not budget-bound relative to each other even though they are not comparable with the headline.

3.35 Four further ways the evaluation reported a number that was not a measurement

Secs. 3.20 and 3.25 record four ways this project’s benchmark harness produced a number that was not a measurement, and Sec. 3.26 a fifth in which a check could not fail. Standing the periodic evaluation up on a live training run produced four more within one day, on a different part of the pipeline. They are recorded together because the distribution across them is the finding: three of the four *refused*, and only the one whose failure was inside the scoring path produced a number.

(vi) **An aborted generation graded as a wrong answer.** An eight-problem run returned a plausible 0.125. Opening the generations shows why: of the eight, **six carry finish_reason**: "abort", one was cut off after 75 characters, and every one of the six was graded `correct`: `false` (`work_pe/dryrun_out/step50/olympiadbench_generations.jsonl`, eight rows, six `abort` and two `stop`, one correct). The cause is that the trainer sends `pause_generation` around every weight update, which drops the requests in flight; a curve built from that reads “the model gets everything wrong” when what happened is that the run interrupted its own evaluation. An abort is now a

failed request, excluded from the denominator and counted, and the same eight problems then return an empty-evaluation status and NaN instead of 0.125. This is the only one of the four that produced a number, and the number was in the plausible range for a genuinely weak arm.

(vii) **A cap above the *server*’s context, which the model’s own config cannot reveal.** Modes (ii) and (iv) are a cap larger than the model’s context, which a guard reading the model’s `config.json` can catch. This one is not that. A0’s Qwen2.5-32B declares `max_position_embeddings: 32768`, so the guard passes, while its rollout server was launched with `--context-length 4096` and rejected every request. Ten consecutive periodic-eval points, steps 50 through 500, refused: every one is `status_code 3` (empty evaluation) with `resolved_adapter: null`, no problems graded and a cost of 0.0088–0.0094 of training throughput (ten `step*/results.json` under `runs/a0_periodic.pinnedurl_run.20260902T175124Z/`). A server flag is not visible in a model’s config, so the clamp has to be against the limit the server publishes about itself.

(viii) **An endpoint that could not be discovered at all.** A later point refused with `status_code 6`, `resolved_adapter: null`, empty `token_budget` and `rows`, at a cost of 6.2×10^{-7} of throughput (`runs/a0_periodic/step200/results.json`, written before the launch that produced Table 28). The discovery path read an attribute that exists on the engine class the deployment does not use, so it had never worked; what had been keeping the evaluation alive was a hardcoded address, and removing the hardcoded address is what exposed it. Recorded because the diagnosis that preceded it was wrong in an instructive way: the pin was blamed, and the pin was correct.

(ix) **A reused tracker run id discarding an entire run’s metrics.** A relaunch reused the experiment’s tracker run id, so the tracker resumed the *previous* run at its last step and answered every commit from the new one with `Tried to log to step N that is less than the current step 509 ...this data will be ignored`. That line appears 212 times in the relaunch’s own tracker output (`.../a0_math/t1/wandb/run-20260902_175715-a0_math_t1_train`), one per logged step: **every metric of that run, training statistics and evaluation points alike, was discarded at the tracker while the run itself was healthy**. The on-disk results were unaffected, which is the only reason the loss is recoverable. Run ids are now minted fresh per launch.

What these four add to the rule of Sec. 3.20. That subsection requires a token budget, a truncation fraction and a failure count beside every score. These four say the same requirement applies one level up, to the *evaluation’s own* status: a point that did not run and a point that scored zero must not be the same value on a curve. Three of these four failures were survivable precisely because the module emits a status code and a NaN rather than a zero, so a broken evaluation and a genuinely inert adapter stayed distinguishable; mode (vi) was not survivable that way, because it scored. The convention we adopt is therefore that every evaluation point carries a status alongside its score, and that a plotted curve shows refusals as gaps rather than omitting them.

3.36 Correction, same day, to Secs. 3.34 and 3.35

Both corrections below were forced by the same fact, which is worth stating before them: the periodic evaluation writes each point to a directory keyed *only* by the training step, so a relaunch overwrites the previous launch’s point at the same step in place. Within half an hour of the two subsections above being written, the running arm reached step 200 and overwrote a file one of them

cited. An artifact that a live run can overwrite is not a citable artifact, and the durable record is the training log, which is rotated rather than replaced.

A fourth point, and the retraction of the trend reading. At step 200 the arm scored 0.1094 on the same 64 problems at the same 2048-token budget, served adapter `a0_math-v201`, 64 of 64 graded with no failures. That is exactly the step-50 value, so the series over four points is 0.1094, 0.1250, 0.1406, 0.1094: it rises for three points and returns to where it started. Sec. 3.34 describes the first three as rising monotonically, and while that sentence is true of those three it invites a reading the fourth point removes, so we retract the framing and keep only the disclaimer that went with it — at $n = 64$ these intervals overlap almost entirely and no trend was claimed. The truncation count is also not constant as stated there: it is 3, 3, 3, 2 of 64. The adapter-version check continues to hold exactly, 51, 101, 151, 201.

The citation in mode (viii), redirected and strengthened. That paragraph cites `runs/a0_periodic/step200` for a `status_code 6` refusal. That path now holds the step-200 point described above, because the running arm overwrote it. The durable artifact for the same failure is the previous launch’s rotated training log, `runs/a0_math/train.log.1788377789`, and it records the failure four times rather than once — at steps 50, 100, 150 and 200, each as a matched pair of lines:

```
periodic eval cannot read the rollout endpoint at step N: RuntimeError: RolloutController
exposes no inference server address, so there is no endpoint to evaluate against.
periodic eval step=N status=6 diagnosis=-1 model=UNRESOLVED trend=nan@max.tokens=UNKNOWN
live=na best=nan@-1 0.0s (0.0% of throughput)
```

Four consecutive refusals, none of them scoring, at no measurable cost to throughput. The substance of mode (viii) is unchanged and its evidence is four times larger than reported.

3.37 The evaluation’s cost growth was a transient, and is capped anyway

The paragraph “The cost of the check, and what does not explain it” in the fifth-failure-mode subsection above reads three cost points as an unbounded rise and projects that the periodic evaluation consumes the whole of training throughput near step 1900. Five further points now exist. **They refute the projection:** the rise was a transient that peaked at step 150, and the series is flat thereafter. The reading is retracted; the remedy recommended there is implemented regardless, as a safety net rather than as a correction to a growth that is happening.

The slope collapses once the transient is out of the window. Fitted to the first three points the benchmark phase rises at +0.667 seconds per training step, which is the slope that produced the step-1900 projection. Fitted to the five points from step 200 onward it is +0.013 seconds per step, and that residual is not resolvable: across its own 200-step window it implies a rise of 2.6 seconds against a point-to-point scatter of 3.1 seconds about the mean of 122.5. In throughput terms those five points are 10.10%, 9.82%, 10.45%, 10.52% and 9.78% — mean 10.13%, standard deviation 0.31 percentage points. The series is flat, and step 150 is the maximum of all eight rather than a point on a trend. The two negative findings the earlier paragraph established hold across all eight points and are strengthened by them: mean generated length stays in [2287, 2397] characters with no trend, truncations stay at 2 to 4 of 64, all 64 problems grade at every point, and the implied training step time moves only between 26.6 and 28.1 seconds. What changed is the conclusion drawn from three points, not the measurements.

Table 29: Cost of one A0 periodic evaluation point, all eight points, read from `runs/a0_periodic/step{50,...,400}/results.json`. *Benchmark* is the generation and grading phase alone; *total* adds the liveness probe and adapter resolution. *Throughput* is the total divided by the 50 training steps it is amortised over. Step time is implied from the two, as $\text{total}/(50 \times \text{throughput})$.

step	benchmark (s)	total (s)	throughput	mean chars	truncated
50	88.7	104.2	7.85%	2289	3
100	123.9	139.5	10.30%	2336	3
150	155.4	177.4	12.95%	2287	3
200	122.9	138.6	10.10%	2323	2
250	117.4	133.1	9.82%	2397	4
300	124.0	139.8	10.45%	2287	2
350	126.7	142.5	10.52%	2382	4
400	121.5	137.2	9.78%	2312	3

We do not replace one extrapolation with another. Fitted to all eight points the slope is +0.034 seconds per step, and extrapolating *that* would cross the ceiling below near step 2200 — inside this run. The post-transient fit extrapolates to step 5400. Neither number is offered as a prediction: the all-eight fit pools a transient with a plateau and estimates neither, and the post-transient slope is smaller than the scatter it is fitted through, so its crossing point is an artefact of a slope that is not distinguishable from zero. The honest statement is the one the earlier paragraph made and this one keeps: a handful of points cannot separate these shapes, and nothing measured so far bounds the cost. What has changed is that the specific alarming projection is refuted, and that the bound is now enforced rather than predicted.

The cap, and what it is a cap on. Before each point the projected cost is computed as the *previous* point’s wall seconds divided by the cadence times the current measured step time; if that fraction exceeds a ceiling the point is skipped, and the skip is emitted and persisted as a point with `status_code 11 (cost_capped)` and NaN for every measurement. The projection is deliberately the last measured cost rather than a fitted trend: a trend fitted to a handful of noisy points is exactly what produced the step-1900 figure this subsection retracts. The ceiling is 0.15, set by `SELFEVO_PERIODIC_EVAL_MAX_THROUGHPUT_FRAC`, chosen to sit above every point in Table 29 — including the step-150 peak — so that it does not silence a curve that is behaving. Two new series, `periodic_eval/cost/projected_throughput_frac` and `periodic_eval/cost/cap`, are emitted on every point, scored and skipped alike, so a gap on the curve can be read against the number that caused it and the ceiling it was measured against. The cadence (50 steps) and the problem limit (64) are unchanged: they set what the curve measures, and the cap is about what it costs.

Three cases deliberately do not skip, and one property is worth stating. The first evaluation of a process is never capped, because there is no previous point to project from; a step time that is not yet measured yields a NaN projection, and every comparison against NaN is false, so the point runs rather than being blinded for a reason that has nothing to do with cost; and a ceiling of zero or less disables the cap outright. The property: while the cap is firing, the previous cost is not refreshed — a skipped point measures nothing — so once tripped the cap latches until the step time rises enough to bring the same projection back under the ceiling. That is intended, and it is safe to read precisely because a skipped point is a recorded point: a latched cap appears as

a run of `status_code 11` on the curve, not as an absence.

The cap does not protect the run that motivated it. The A0 arm now training imported this module at launch, before the change, so it continues under the uncapped code and the cap takes effect at the next launch. This is stated rather than glossed because the two facts are easy to combine into a false one: the running arm is not protected by the cap, and on the evidence of Table 29 it does not need to be.

The controls the cap is held in place by. Twelve mutations of the cap, applied one at a time to a copy of the tree, MUT_SUMMARY Among them: a projection hard-wired to zero so nothing is ever capped; the ceiling comparison reversed; a skipped point reporting the success status instead of its own; the decision computed and then ignored; the hook never recording what a point cost, so the cap can never fire; an unmeasurable step time skipping rather than running; the ceiling read from the wrong configuration field; the cadence dropped from the projection; the default ceiling silently changed; the projection not recorded on the skipped point; the skipped point never written to the artifact directory; and a skipped point emitting zeroes where it should emit NaN. The last of these is the mutation that matters most for how the curve reads, since a zero `trend_score` is indistinguishable from a model that got every problem wrong — the failure mode Sec. 3.35 is about.

A stale run’s checkpoints were separated from the live run’s, and the separation raced the saver. The saver directory for the A0 arm mixed two runs. The trial name `a0_math/t1` has been reused across seven launches, so all seven wrote checkpoints into `areal-runs/checkpoints/ubuntu/a0_math/t1/default/` keyed only by step — exactly the overwrite-in-place shape Sec. 3.36 records for the evaluation artifacts, and with the same consequence: any retrospective scoring of “A0’s checkpoints by step” would have mixed two runs silently, and the highest-numbered checkpoints would have come from a run that no longer exists. The live run was near step 350; the same directory also held steps 374, 399, 424, 449, 474 and 499, whose directories all carried modification times from 16:39 to 17:38 UTC, inside the window of the launch that was killed at step 509. Those six were moved — renamed within the same filesystem, never deleted — to `areal-runs/stale_checkpoints/a0_math_t1_run_killed_at_step509_20260902T1738Z/`, and every file matched its pre-move size and MD5 at the destination.

One of the six was not stale, and the check that caught it is not the check that verified the move. Between the survey that classified the six at 22:51:20 and the checksum pass some tens of seconds later, the live run reached step 374 and its saver overwrote that path in place at 22:52:00. The move therefore carried the *live* run’s fresh step-374 checkpoint out with the five genuinely stale ones. The before/after checksums agree — they were both taken after the overwrite, so they verified the move faithfully and could not possibly have caught the mislabelling. What caught it was the *file* modification time: 22:52:00 for step 374 against 16:51 to 17:38 for the other five. A directory’s modification time is not evidence about the files inside it, and a survey taken a minute before a move is not evidence about the state at the moment of the move; against a live writer, provenance has to be re-established at the instant of the operation, not inherited from a listing. Step 374 has been moved out of the stale directory to `areal-runs/misplaced_live_checkpoint_step374/`, which records the account and the single command that returns it to the saver directory; until that is run the live run’s saves have a hole at step 374 and nothing else is affected.

What was already lost, and what is not ambiguous. No surviving checkpoint is ambiguous: every one left in the live directory has a file modification time after the live launch began at 19:43. The losses had already happened and are not recoverable. The killed run’s checkpoints for steps 24 through 349 were overwritten in place by the two launches that followed it, before any of this; its step 374 was overwritten by the live run at 22:52:00 during the separation. Steps 399, 424, 449, 474 and 499 are the only surviving weights from that run, and the margin was minutes rather than hours: the move completed at 22:52, and the live run wrote its own step 399 into the freed slot at 23:03:35 and its step 424 at 23:17:26. Both of those stale checkpoints would already have been destroyed had they been left in place, and the remaining three were inside the following hour. That is what makes the separation a rescue rather than a tidy-up — and it is also why the race in the preceding paragraph happened at all.

3.38 The A0 baseline is flat at the headline budget too, and the cheap probe was wrong about the level but right about the shape

Sec. 3.34 and Sec. 3.36 leave one question open, and it is the question that decides what to do next: the periodic series is flat around 0.11, but every point in it was taken at 2048 new tokens on 64 problems, so the flatness could be the run failing to learn or it could be an instrument too blunt to see learning. Those two readings imply opposite next moves. This subsection settles it by scoring the saved checkpoints at the *headline* budget on a problem set large enough to resolve a difference worth acting on, with the un-adapted base model included as the control. **The answer is that both halves of the worry are partly right, and they do not cancel: the cheap probe understates the accuracy by about 32 points, and its flatness is nonetheless real.**

What was run. Every arm scores all 338 problems of the committed OlympiadBench `search` half at 16,384 new tokens, greedy, one sample — the headline protocol exactly, with `max.tokens` taken from `BENCH_OVERRIDES` rather than passed on the command line, so it is the same constant the periodic evaluation reads. Thirty-two evaluations were run on eight B200s: six replicates of the base model, the 22 checkpoints from step 24 to step 549, and four repeat measurements described below. All 32 graded 338 of 338 with *zero* request failures, and truncation never exceeded 5 of 338, so no score here is budget-bound — unlike the periodic series, the cap is not what is being measured. The `report` half was deliberately not touched: it is the reserved half that makes the headline claim meaningful, and spending it on a diagnostic sweep would have destroyed the search/report discipline for a question that 338 problems can answer.

Every arm is served by an identical code path, on purpose. Each LoRA was merged into the base weights and served as a standalone model, so the base control and the checkpoints differ only in the weights and not in whether an adapter kernel ran at all. A base scored without `--lora-paths` against checkpoints scored with it would confound the one comparison this subsection exists to make.

The checkpoints are provably from the live run. That directory held a killed run’s checkpoints until they were separated by hand, so identity was re-established from the weights rather than from the path. The Frobenius norm of the merged LoRA delta rises strictly monotonically across all 22 checkpoints, 0.0758 at step 24 to 3.4858 at step 549 with no break and exactly 256 merged tensors every time; file modification times are strictly increasing and all fall after the live launch at 19:43; every adapter file is 134,287,464 bytes. A checkpoint from a different run does not lie on another run’s trajectory, so the monotone series is the check that the size and the path cannot provide. The

same trajectory answers a second question at no extra cost: the adapter is not inert. It grows by a factor of 46 over the run, reaching a relative perturbation of 2.46×10^{-3} , so whatever is wrong is not that the optimiser failed to move the weights.

The instrument’s own noise band, measured rather than assumed. The base model was scored six times from six separate servers. At temperature 0 on fixed weights the only remaining source of variation is batching nondeterminism, so the spread across those six is what this harness returns when *nothing* is different. It is not small: 0.4083, 0.4320, 0.4379, 0.4379, 0.4231, 0.4290 — mean 0.4280, standard deviation 0.0112, range 3.0 points, with 14 to 22 of 338 problems changing answer between two runs of identical weights. **Three of the six same-weights pairs differ significantly by McNemar:** `base_b` – `base_a` = +0.0237 with 95% CI [+0.0020, +0.0454], and both `base_c` and `base_d` against `base_a` give +0.0296 with 95% CI [+0.0024, +0.0568]. This is a sixth entry for the catalogue of Sec. 3.20: McNemar’s interval models sampling error only, and greedy decoding on a batching server carries a second source it does not model, so a single paired comparison against a single base measurement will report a significant gain between two copies of the same weights at a rate far above 5%. A base control has to be *replicated*, not merely present.

The curve. Table 30 gives it. The 22 checkpoints span 0.4142 to 0.4615, a range of 4.7 points, against a same-weights band of 3.0 points. Pooled over all 22, the checkpoints average 0.4369 against the base’s 0.4280, a difference of +0.89 points with a bootstrap 95% CI of [−0.56, +2.39] — consistent with zero, and tight enough to have resolved a true effect of 2.4 points. Fitting accuracy against step gives a slope of +0.13 points per 100 steps, 95% CI [−0.21, +0.49]; over the 525 steps measured that is +0.68 points with 95% CI [−1.09, +2.56]. Both intervals are computed by resampling *problems*, which is what the arms share, rather than resampling arms.

The three points that looked significant do not replicate. Three checkpoints cleared $|z| > 2$ against the base mean — steps 199, 499 and 549. At 22 comparisons that is close to what $\alpha = 0.05$ produces by chance, and none survives a Bonferroni threshold of $|z| > 2.86$, but the decisive test is to measure them again rather than to argue about multiplicity. Each was re-scored on a fresh server, together with the lowest checkpoint as a control on the direction of the effect. Every one moved back toward the base: step 199 fell from 0.4615 to 0.4320, step 549 from 0.4615 to 0.4438, step 499 from 0.4527 to 0.4467, while step 24 rose from 0.4142 to 0.4231. Averaged over both measurements no checkpoint sits more than 2.5 points above the base mean, which is inside the same-weights band. **The apparent significance was the batching noise of the preceding paragraph, caught by the control it predicts.**

What this supports, stated plainly. At the headline budget, across 525 training steps, the A0 baseline is statistically indistinguishable from the un-adapted base model. The design would have resolved a pooled effect of 2.4 points or a drift of 2.6 points over the run and it found +0.89 and +0.68 respectively, both with intervals containing zero. So the reading is *not* that the cheap probe was too blunt to see a real gain: at seven times the sensitivity and eight times the budget there is still no gain to see. What the cheap probe did get badly wrong is the level — 0.11 against 0.43, a 32-point understatement — which is exactly the distortion Sec. 3.34 built its naming convention to prevent, now measured on this run rather than argued from the general case. Combined with the weight-space trajectory, which shows the adapter growing by $46\times$ throughout, the diagnosis is the “learning but not helping” branch of `periodic_eval.diagnose`, not the “not learning” branch:

Table 30: A0 at the headline budget: OlympiadBench **search** half, all 338 problems, 16,384 new tokens, greedy, one sample. **base** is the mean of six replicates of the un-adapted model, whose own range is 3.0 points. Δ is against that mean and z is Δ over its bootstrap standard error. Every row graded 338 of 338 with no request failures. **Unlike Table 28 these are accuracies at the headline budget and are directly comparable with the rest of this paper.**

step	accuracy	Wilson 95%	Δ vs base	z
base ($n = 6$)	0.4280	—	—	—
24	0.4142	[0.363, 0.467]	−0.0138	−1.03
49	0.4467	[0.395, 0.500]	+0.0187	+1.41
74	0.4231	[0.372, 0.476]	−0.0049	−0.35
99	0.4497	[0.398, 0.503]	+0.0217	+1.75
124	0.4379	[0.386, 0.491]	+0.0099	+0.77
149	0.4349	[0.383, 0.488]	+0.0069	+0.46
174	0.4527	[0.400, 0.506]	+0.0247	+1.74
199	0.4615	[0.409, 0.515]	+0.0335	+2.48
224	0.4290	[0.377, 0.482]	+0.0010	+0.07
249	0.4408	[0.389, 0.494]	+0.0128	+0.91
274	0.4290	[0.377, 0.482]	+0.0010	+0.08
299	0.4260	[0.374, 0.479]	−0.0020	−0.15
324	0.4172	[0.366, 0.470]	−0.0108	−0.85
349	0.4349	[0.383, 0.488]	+0.0069	+0.48
374	0.4379	[0.386, 0.491]	+0.0099	+0.83
399	0.4231	[0.372, 0.476]	−0.0049	−0.41
424	0.4349	[0.383, 0.488]	+0.0069	+0.49
449	0.4290	[0.377, 0.482]	+0.0010	+0.08
474	0.4349	[0.383, 0.488]	+0.0069	+0.51
499	0.4527	[0.400, 0.506]	+0.0247	+2.10
524	0.4408	[0.389, 0.494]	+0.0128	+0.87
549	0.4615	[0.409, 0.515]	+0.0335	+2.46

the optimiser is moving the weights steadily and none of that movement reaches OlympiadBench accuracy.

What this does not support, and what would settle it. This is 549 steps of a run configured for 12,853, so it bounds the gain *so far* and not the gain the run can eventually produce; a null at 4% of a schedule is not a null at the end of one. It is also one benchmark, and a reward that is teaching something OlympiadBench does not test would look exactly like this. Two things would separate those cases. First, the same sweep on the training reward itself: if the reward is rising while accuracy is flat, the gap is the claim and it is measurable with the artifacts already saved. Second, continuing this curve as the run proceeds — the sweep costs about ten GPU-minutes per checkpoint and the harness, merge and analysis are stored beside the scores in `gs://selfevo/evals/a0_headline/`, so extending it is re-running one script rather than rebuilding the measurement.

3.39 The interference null survives a second run, a later checkpoint, and a swept random control

Sec. 3.28 closed the pre-registered gate on one checkpoint. Three objections survive that measurement and each of them, if right, would reopen the method: that the batch was favourable, that the

effect appears only once the policy has specialised, and that a single draw of the size-matched random partition is not a null distribution. This subsection answers all three, and adds a provenance correction to Sec. 3.28 that has to be recorded because the checkpoint it names no longer holds the weights it was measured on.

Provenance correction to Sec. 3.28. The dump behind Table 23 read the adapter at `a0_math/t1/default/epoch0epochstep199globalstep199` at 09:41 on 2026-09-02. The file now at that path was written at 21:22 the same day by the currently live run, which reuses the same experiment and trial names and therefore the same directory. The tensor Table 23 measured is consequently *not* the tensor at that path, and that table cannot be re-derived from the checkpoint directory as it now stands. Nothing in the result changes — the dump, its 496 exact gradient pairs and its metadata are all still on disk — but the identity has to be carried by recorded checksums rather than by a path, and every measurement below records one.

The batch is the most readable one available, not a favourable one. A GRPO group whose rollouts all score alike carries an identically zero advantage and therefore an exactly zero gradient, so a batch of mostly unanimous groups drives every partition’s mean cosine toward the same value for a reason that has nothing to do with clustering. That is the false-negative this gate was pre-registered to refuse, and the first dump taken on this project hit it: 90 of 128 groups were unanimous. The batch used here and in Sec. 3.28 (`a0_step199_probe.jsonl`, md5 078a2ba0541a2a19b9a9c95449062ad6) carries 140 groups at group size 8 with k -histogram {1:20, 2:19, 3:20, 4:14, 5:15, 6:17, 7:35}: **140 of 140 groups are non-unanimous**, against a pre-registered requirement of 60%. The dumps confirm it downstream — zero zero-valued GRPO sketches in every dump reported here. Whatever this null is, it is not the sparse-advantage artefact.

A second run, never previously analysed. A readable dump also exists at `globalstep49` of the `harnessT_trunc` experiment — a different run, same base (Qwen2.5-32B-Instruct), same adapter geometry ($r=32$, $\alpha=32$, on q/k/v/o) — over 153 groups, again 153/153 non-unanimous, k -histogram {1:28, 2:24, 3:19, 4:11, 5:16, 6:18, 7:37}. It had been dumped but never analysed. Table 31 gives it. The contrast changes sign across the three settings (+0.0029, −0.0004, +0.0263) and no setting resolves: the largest is still below the 0.0331 sketch resolution floor. The `mcs = 5` row is reported for completeness but does not meet the pre-registered $K \geq 4$ requirement — it finds three clusters, which is one pair per partition.

What was run for the trajectory. Three adapters from the *live* A0 run were re-dumped on eight H200s, on the identical batch, with the identical sketch (`dim 8192`, `seed 0`) and the identical library versions as the H100 dump (torch 2.9.1+cu128, transformers 5.3.0, peft 0.18.1); the repository’s own 175 cluster-LoRA tests pass on that box, including the test that pins the `hidden_states` layout the MEDS feature depends on. Identity is established the way Sec. 3.38 established it, because the path cannot carry it: md5 b8319909729618a04f3f5a90faf9468b at step 199, a876c1c068274251a4001f35e8c261c9 at step 449 and 65aa994558e845bea9b3bf63b720ac70 at step 749, each 134,287,464 bytes with exactly 256 merged tensors, and Frobenius norms of the merged LoRA delta of 1.2309, 3.0912 and 4.7495 — strictly increasing, so the three lie on one run’s trajectory and the adapter is demonstrably not inert across the span being tested.

One caveat, stated rather than buried. The rollouts are held fixed across the three checkpoints, so at steps 449 and 749 they are off-policy and the stored rewards, and hence the advantages, are

Table 31: A second run, never previously analysed: mean pairwise cosine between per-cluster gradients at `globalstep49` of `harnessT_trunc`, 153 groups, 153/153 non-unanimous. The hypothesis requires *behavioural* to sit below *random*. The contrast changes sign across settings and no row resolves against the 0.0331 floor. The `mcs = 5` row finds three clusters and so does not meet the pre-registered $K \geq 4$.

min. cluster size	clusters	behavioural	random	prompt-grad.	behav. – random
2	30	+0.00282	−0.00007	+0.00069	+0.00289
3	15	+0.00719	+0.00759	−0.00124	−0.00040
5	3	+0.01375	−0.01253	−0.02562	+0.02628

Table 32: The interference null along one run’s trajectory. Three checkpoints of the live A0 run (identity established by the strictly increasing merged-LoRA norms 1.2309, 3.0912, 4.7495), the same 140-group batch, the same sketch. *behav.* and *random* are mean pairwise cosines between per-cluster gradients; the random control is redrawn at five seeds and z is the behavioural-minus-random contrast in units of that control’s own spread. The hypothesis requires the contrast to be *negative* and large. Nothing resolves against the 0.0331 sketch floor at any checkpoint, the sign depends on the clustering setting rather than on the checkpoint, and the contrast does not grow as the policy specialises.

step	min. clust.	K	behav.	random	behav. – random	z
199	2	25	−0.00151	+0.00019	−0.00170	1.19
199	3	17	−0.00110	−0.00071	−0.00039	0.22
199	5	8	+0.00423	−0.00102	+0.00526	1.34
449	2	26	−0.00090	+0.00049	−0.00139	0.89
449	3	18	−0.00079	+0.00002	−0.00081	0.85
749	2	24	−0.00169	+0.00059	−0.00228	1.17
749	3	18	−0.00054	−0.00001	−0.00053	0.49
749	5	7	+0.00742	+0.00119	+0.00624	1.48

those of the earlier policy. That is deliberate: holding the batch fixed is what makes the comparison across checkpoints a statement about the *parameters* rather than about two things moving at once. What it cannot rule out is an interference structure that only appears in a batch drawn from the later policy itself.

Why the random control is swept. Sec. 3.28 compares the behavioural partition against *one* draw of a size-matched random partition. A single draw is a point, not a null distribution, and the quantity that decides the gate is whether the behavioural partition sits outside the range that feature-blind labels of the same sizes produce. We therefore redraw the control at five seeds per setting and read the contrast against its own spread, which is the comparison the pre-registered rule was reaching for.

The verdict, and what it costs. Across four checkpoints from two runs, three clustering settings and five control seeds, the behavioural partition never separates gradients from a size-matched random one by more than about 1.5 standard deviations of the control’s own spread, and every contrast sits an order of magnitude inside the sketch’s resolution floor. At the settings that meet the pre-registered $K \geq 4$ with the most clusters the sign is nominally favourable and never significant; at the coarsest setting it is unfavourable. **The pre-registered rule closes the gate: the clusters**

are not a gradient-separating structure, and no per-cluster adapters are built on this signal. The one escape the plan had reserved — that interference grows as the policy specialises — is measured here across a 550-step span and does not occur.

3.40 Per-prompt credit has a prerequisite that the corpus, not the controller, decides

Per-prompt credit assignment pairs a routing decision with the *next sighting of the same prompt*, so the interval at which any credit becomes available is a property of the data loader:

$$\text{recurrence period} = \frac{n_{\text{prompts}}}{\text{batch size}} \text{ steps.} \quad (7)$$

This is arithmetic, and it disqualifies the configuration our own baseline runs in. A0 trains on 102,835 decontaminated DeepMath rows at a batch of 8, so its epoch is 12,853 steps and it is presently at step 852 of it. A learned-router arm launched on that corpus would emit thousands of routing decisions and receive **zero** credits for the entire run, while every logged quantity except `prompt_credit/credited` reported it healthy. That is not a null result, it is a no-op that reads as one, and it is invisible in simulation because a simulated world recurs by construction.

We therefore state it as a precondition rather than a detail: *a per-unit credit rule across time is defined only on a corpus the run revisits*, and the revisit rate, not the estimator, sets how much signal exists. A second-order term follows from the self-baseline rule, under which a prompt’s first delta is withheld because a zero baseline would hand the first-credited mode the whole common training trend: the first credit therefore arrives at a prompt’s *third* sighting, after two recurrence periods.

Our M8 arms consequently train on a seeded 512-row subset of the same decontaminated corpus, giving a 64-step epoch — confirmed in the run header (Epoch 1/10 Step 1/64) and in the ledger, whose `pending` count rises by exactly 8 per step to 512 at step 64 with zero evictions and zero same-batch skips. The subset size is a *consequence of Eq. 7*, not a tuned hyper-parameter, and it is the arms’ principal limitation: they say what a learned router does when credit is available, not what it does at A0’s corpus size, where by Eq. 7 it is not.

3.41 What carries the routing claim, and the instrument checked against two controls

The mode mix is not evidence. In a controlled world where a right answer exists, a batch-credited router’s L_1 distance from uniform *rises* while its targeting stays at chance, because the router chooses its own assignment and the arms drift apart on that feedback alone. Results are therefore carried by **subset contrast** — half the total variation distance between the mode distribution on one subset of groups and on another, which is 0 for a router with a favourite mode however lopsided the mix becomes — with direction checked by a **per-subset targeting fraction**, so an inverted preference cannot pass.

Two subset labellings are reported and they answer different questions. *Silence* is the fixed rule’s own partition (solved-silent, unsolved-silent, informative); a router that separates these is doing what the hand-written rule already does, so this is the floor and not the finding, and its targeting fraction is reported as *agreement with the rule*, never as correctness — a live run has no ground truth about which mode helps a prompt. *Covariate* is a median split on one covariate *within* the informative stratum, which holds the pass count approximately fixed, so any contrast there is attributable to something the pass count does not carry. That is the axis our own feature audit records as untested rather than refuted.

Table 33: The measurement instrument against a positive and a null control, 200 steps through the real `_compute_advantages`. Brackets are 95% bootstrap intervals over steps.

router	contrast on <code>mean_logprob</code>	on <code>logprob_dispersion</code>	silence contrast
reads <code>mean_logprob</code> only	0.9882 [0.831, 1.000]	0.2071	0.0098
feature-blind (null)	0.0616 [0.020, 0.228]	0.0440	0.0528
difference	+0.9266 [+0.670, +0.954]	+0.1632	−0.0430

Table 34: `credit_sim` re-run on the tree that now exposes the rule from configuration. Final-quarter subset contrast, mean over 8 paired seeds, with the error bar on the *difference* and the number of seeds in which it has the reported sign.

arm	final contrast	difference	value
<code>batch</code>	0.098	<code>prompt</code> − <code>batch</code>	+0.682 ± 0.021 (8/8)
<code>prompt</code>	0.779	<code>prompt</code> − shuffled	+0.678 ± 0.018 (8/8)
<code>prompt</code> , credit shuffled	0.102	self-baseline − shuffled	+0.685 ± 0.024 (8/8)
<code>prompt_self_baseline</code>	0.828	self-baseline − <code>prompt</code>	+0.048 ± 0.012 (7/8)
same, credit shuffled	0.143	<code>prompt_centered</code> − <code>prompt</code>	−0.027 ± 0.010 (1/8)
<code>prompt_centered</code>	0.752		

Every number is a difference against a control, bootstrapped over *steps* rather than groups because within one step the decisions share a policy, a batch and a parameter vector. The two arms are resampled independently: rollout generation is not deterministic across inference servers, so two arms with the same seed and the same data order see different completions from step one and are independent draws from one configuration, not a matched pair.

Before any arm was read, the statistic was driven through the real advantage path for 200 steps against a router that reads one covariate and nothing else, and against a feature-blind router drawing modes from fixed proportions (Table 33). It fires at 0.99 on the covariate a router actually used and reports 0.06 for a router that used nothing, so a contrast reported below is not a property of the estimator. A covariate that is constant within a window is reported *undefined*, never 0.

3.42 The credit study reproduces exactly with the config path in place

Making `credit="prompt_self_baseline"` selectable from configuration required editing the two lines in the actor and the config dataclass that the CPU study had deliberately left alone. Table 34 re-runs that study on the edited tree: every figure it is cited by is recovered to three decimals, over the same eight paired seeds.

The last row is a negative result and is reported as one. `prompt_centered` subtracts the batch’s mean delta, which is one number shared by every arm — the same class of quantity per-prompt credit exists to escape — and it measures *below* plain per-prompt credit. It remains selectable because it is the rung of the ablation that shows the distinction matters, not because it is a candidate.

3.43 The learned router on a GPU: arms, and the pre-credit window as a null check

Four arms train concurrently on 8×B200, two GPUs each (`fsdp:d1p1t1` actor + `sglang:d1p1t1` rollout; the fp32 32B actor fits one card unsharded at a measured 133.7 GB of 178.35, so there is exactly one router per arm rather than one per data-parallel worker). Everything except the routing

Table 35: Pre-credit window, steps $[0, 20)$ on every arm — 160 routing decisions per arm, 0 credits in either. Windows are matched on *step index*, i.e. policy age, because the arms were launched at different times. Brackets are 95% bootstrap intervals over steps; the two arms are resampled independently, since rollout generation is not deterministic across inference servers and the arms are independent draws from one configuration rather than a matched pair.

subset contrast	treatment	matched random	difference
silence (solved vs unsolved)	0.3966	0.2857	+0.1108 $[-0.072, +0.260]$
cov. mean_response_len	0.0864	0.0732	+0.0132 $[-0.230, +0.229]$
cov. len_dispersion	0.0682	0.1463	-0.0782 $[-0.259, +0.163]$
cov. mean_logprob	0.1397	0.0976	+0.0421 $[-0.191, +0.291]$
cov. logprob_dispersion	0.1383	0.0976	+0.0408 $[-0.177, +0.243]$
cov. truncated_fraction	undefined — constant within the informative stratum		

configuration is A0’s: Qwen2.5-32B-Instruct, LoRA $r=32$ on q/k/v/o, 8 prompts $\times$ 8 samples, cap 1024, lr = 10^{-4} , kl_ctl = 0, adv_norm = null.

arm	router	credit	role
m8_prompt	contextual	prompt_self_baseline	treatment
m8_control	random	(inert: no observe)	rate-matched control
m8_shuffle	contextual	prompt_self_baseline + shuffle	correspondence control
m8_batch	contextual	batch	the recorded null, re-run

The control replays the treatment’s *measured* proportions, taken from the treatment’s own route/{mode}_groups over its first 19 steps (rl = 0.3882, sft = 0.2961, skip = 0.3158) and supplied through two independent paths — the environment the router reads and a file the preflight compares it against — so the configuration cannot agree with the expectation by construction. Over the window below the control realised 0.325/0.306/0.369 against the treatment’s own 0.400/0.288/0.313, inside three binomial standard errors at this sample size.

A pre-registered null, and it holds. By Eq. 7 no credit exists before a prompt’s third sighting, i.e. step ≈ 128 here, and over the window reported in Table 35 the ledger’s **credited** count is 0 for every step of every arm while **pending** rises by exactly 8 per step. The learned router has therefore received no feedback and its ridge models sit at initialisation for every mode, so it must *not* separate itself from a router that reads nothing. It does not: every difference against the matched control contains zero. This window is reported because it is the check that says the instrument and the control are not manufacturing a result — it is not the claim, and the window where the claim lives is stated below as still open.

What is still open, stated as such. These arms establish that the learned router is selectable, runs at 32B beside its mandatory control, and does *not* beat that control before it is taught anything. They do not yet say whether it beats it once credit flows: the post-credit window $[128, \cdot)$ opens only after two recurrence periods and is being written continuously to `gs://selfevo/runs/b200/_analysis/`, together with the correspondence control that shuffles credits across the prompts that earned them and the batch-credited arm that the identity predicts cannot learn at all. A difference reported there, and only there, would be the first evidence that a learned router decides better than the fixed rule in training.

Eq. 7 verified live, step for step. The treatment arm’s ledger behaves exactly as the arithmetic requires on the running 32B job. `prompt_credit/pending` rises by precisely 8 per step and saturates at 512, the corpus size, with zero evictions and zero same-batch skips. `prompt_credit/cold_baseline_skips` becomes non-zero at *exactly* step 64 — one recurrence period — which is the first delta being withheld because a prompt has no earlier delta to centre against. The *first credit* lands at *exactly* step 128, two recurrence periods, and the credited values are genuinely per-unit rather than a shared scalar: over the first 32 they span $[-0.625, +0.750]$ with standard deviation 0.250 around a mean of -0.020 . The interval between a decision and the credit that closes it is not the epoch length but scatters around it (45 steps for the first), because prompts are reshuffled within an epoch. This is the first time a learned router in this project has been credited with anything that varies by unit, and it is the precondition every result in the post-credit window rests on.

Reproducibility notes

Every number in Sec. 3 names the run that produced it. Benchmarks are scored with the same grader across arms; the grader’s own bias was measured and corrected once (an earlier version discarded unbalanced final boxes, which penalised degraded checkpoints specifically). Intervals are reported on differences, paired by item, never as independent per-arm intervals.